



# TESTING FOR THE VERIFICATION OF COMPLIANCE OF PV INVERTER WITH:

## UL STANDARD FOR SAFETY FOR PHOTOVOLTAIC (PV) DC ARC-FAULT CIRCUIT PROTECTION, UL 1699B

REVISION, DATED MAY 18, 2021

Test Report Number.....: GZES240100172701

Trademark.....:   
MEGAREVO

Tested Model.....: R12KLNA-P6

Variant Models.....: R8KLNA-P6-P4, R12KLNA-P4, R16KLNA-P4,  
R8KLNA-P6-P6, R16KLNA-P6

### APPLICANT

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Address.....: Building 2, Winlead, Shutianpu Community, Matian Street,  
Guangming District, Shenzhen, Guangdong, China

### TESTING LABORATORY

Name .....: SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou  
Branch

Address.....: 198 Kezhu Road, Science City, Economic & Technology  
Development Area, Guangzhou, Guangdong, China

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(Technical Reviewer)

Date of issue .....: 2024-09-04

Number of pages .....: 45



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### Test Report Historical Revision:

| Test Report Version | Date       | Resume         |
|---------------------|------------|----------------|
| GZES240100172701    | 2024-09-04 | First issuance |

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## 1 SCOPE

SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch has been contract by Shenzhen Megarevo Technology Co., Ltd. in order to perform the testing according to following standards:

- Clause 25, 27.2, 27.3, 28, 29, 30, 31, 35, 42, 55, and 56 of UL Standard for Safety for Photovoltaic (PV) DC Arc-Fault Circuit Protection, UL 1699B, Revision, Dated May 18, 2021.

## 2 GENERAL INFORMATION

### 2.1 Testing Period and Climatic conditions

The necessary testing has been performed between the 22th Feb. 2024 and the 13th Apr. 2024.  
All the tests and checks have been performed in accordance with the reference Standard (the tests are done at  $25 \pm 5^{\circ}\text{C}$ ,  $96 \text{ kPa} \pm 10 \text{ kPa}$  and  $65\% \text{ RH} \pm 10\% \text{ RH}$ ).

#### SITE TEST

Name ..... Dongguan BALUN Testing Technology Co., Ltd.  
Address..... Room 104, 204, 205, Building 1, No. 6, Industrial South Road, Songshan Lake District, Dongguan, Guangdong, China

### 2.2 Equipment under Testing

Apparatus type/ Installation ..... Hybrid Solar Inverter / Fixed installation  
Manufacturer/ Supplier/ Installer..... Shenzhen Megarevo Technology Co., Ltd.  
Fixed (permanent connection)  
Trademark.....   
Model/ Type ..... **R12KLNA-P6**  
Serial Number ..... M10824400001  
Software Version..... Ver 1.0406(DSP), Ver 2.0306(ARM)  
Rated Characteristics ..... **R12KLNA-P6**  
PV side: 120-430V(500Vmax), 16A\*4, 22A;  
Batteries side: 40-58V, 250A/260Amax  
AC grid side: 120Vac/240Vac Split phase, 60Hz, 50.0A(55.0Amax),  
-0.8~+0.8  
EPS<sup>(\*)</sup> side: 120Vac/240Vac, Split phase, 60Hz, 12KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax)  
**(\*)EPS:Equipment or facilities that deliver electric power to a load**

Date of manufacturing: 2023

#### Test item particulars

Input ..... DC  
Output ..... AC  
Class of protection against electric shock ..... Class I  
Cooling group..... Fans  
Modular ..... No  
Internal Transformer ..... Transformer less

## Copy of marking plate (representative):


**MEGAREVO**  
 Hybrid Solar Inverter R8KLNA-P4  
 Grid Support Interactive Inverter
**PV Input**

Max PV Input 12kW

Mppt input Voltage 120~430V

MAX.Input Current 16A\*4

MAX.Short Circuit Current 22A\*4

**AC Output (On Grid)**

Rated Output Power 8kVA

Rated Output current 33.3A

Grid Voltage 120V/240V Split phase

Grid Frequency(Optional) 60Hz

Power Factor Range -0.8~+0.8

Max.Grid Passthrough Current 200A

**AC Output (EPS)**

Rated Output Power 8kVA

Rated Output Current 33.3A

EPS Voltage 120V/240V Split phase

EPS Frequency(Optional) 60Hz

**Battery**

Battery Voltage Range 40V~58V

Max. charging Current 170A

Max. Discharging Current 170A

**System**

Protection Degree: IP65/NEMA 3R

Dimensions(W\*D\*H) 495\*900\*260mm

Weight 55Kg

Max. Efficiency 96.5%

PV DC AFCI Type 1

PVRSS Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


**MEGAREVO**  
 Hybrid Solar Inverter R12KLNA-P4  
 Grid Support Interactive Inverter
**PV Input**

Max PV Input 18kW

Mppt input Voltage 120~430V

MAX.Input Current 16A\*4

MAX.Short Circuit Current 22A\*4

**AC Output (On Grid)**

Rated Output Power 12kVA

Rated Output current 50A

Grid Voltage 120V/240V Split phase

Grid Frequency(Optional) 60Hz

Power Factor Range -0.8~+0.8

Max.Grid Passthrough Current 200A

**AC Output (EPS)**

Rated Output Power 12kVA

Rated Output Current 50A

EPS Voltage 120V/240V Split phase

EPS Frequency(Optional) 60Hz

**Battery**

Battery Voltage Range 40V~58V

Max. charging Current 250A

Max. Discharging Current 260A

**System**

Protection Degree: IP65/NEMA 3R

Dimensions(W\*D\*H) 495\*900\*260mm

Weight 55Kg

Max. Efficiency 96.5%

PV DC AFCI Type 1

PVRSS Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


**MEGAREVO**  
 Hybrid Solar Inverter R16KLNA-P4  
 Grid Support Interactive Inverter
**PV Input**

Max PV Input 24kW

Mppt input Voltage 120~430V

MAX.Input Current 20A\*4

MAX.Short Circuit Current 25A\*4

**AC Output (On Grid)**

Rated Output Power 16kVA

Rated Output current 66.7A

Grid Voltage 120V/240V Split phase

Grid Frequency(Optional) 60Hz

Power Factor Range -0.8~+0.8

Max.Grid Passthrough Current 200A

**AC Output (EPS)**

Rated Output Power 13kVA

Rated Output Current 54.2A

EPS Voltage 120V/240V Split phase

EPS Frequency(Optional) 60Hz

**Battery**

Battery Voltage Range 40V~58V

Max. charging Current 260A

Max. Discharging Current 280A

**System**

Protection Degree: IP65/NEMA 3R

Dimensions(W\*D\*H) 495\*900\*260mm

Weight 55Kg

Max. Efficiency 96.5%

PV DC AFCI Type 1

PVRSS Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

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**MEGAREVO****Hybrid Solar Inverter R8KLNA-P6**  
**Grid Support Interactive Inverter****PV Input**

|                           |          |
|---------------------------|----------|
| Max PV Input              | 12kW     |
| Mppt input Voltage        | 120~430V |
| MAX.Input Current         | 16A*4    |
| MAX.Short Circuit Current | 22A*4    |

**AC Output (On Grid)**

|                              |                       |
|------------------------------|-----------------------|
| Rated Output Power           | 8kVA                  |
| Rated Output current         | 33.3A                 |
| Grid Voltage                 | 120V/240V Split phase |
| Grid Frequency(Optional)     | 60Hz                  |
| Power Factor Range           | -0.8~+0.8             |
| Max.Grid Passthrough Current | 200A                  |

**AC Output (EPS)**

|                         |                       |
|-------------------------|-----------------------|
| Rated Output Power      | 8kVA                  |
| Rated Output Current    | 33.3A                 |
| EPS Voltage             | 120V/240V Split phase |
| EPS Frequency(Optional) | 60Hz                  |

**Battery**

|                          |         |
|--------------------------|---------|
| Battery Voltage Range    | 40V~58V |
| Max. charging Current    | 170A    |
| Max. Discharging Current | 170A    |

**System**

|                    |                             |
|--------------------|-----------------------------|
| Protection Degree: | IP65/NEMA 3R                |
| Dimensions(W*D*H)  | 495*900*260mm               |
| Weight             | 56Kg                        |
| Max. Efficiency    | 96.5%                       |
| PV DC AFCI         | Type 1                      |
| PVRSS              | Tigo RSS transmitters (RS2) |

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

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**MEGAREVO**  
Hybrid Solar Inverter R12KLNA-P6  
Grid Support Interactive Inverter

**PV Input**

|                           |          |
|---------------------------|----------|
| Max PV Input              | 18kW     |
| Mppt input Voltage        | 120~430V |
| MAX.Input Current         | 16A*4    |
| MAX.Short Circuit Current | 22A*4    |

**AC Output (On Grid)**

|                          |                       |
|--------------------------|-----------------------|
| Rated Output Power       | 12kVA                 |
| Rated Output current     | 50A                   |
| Grid Voltage             | 120V/240V Split phase |
| Grid Frequency(Optional) | 60Hz                  |
| Power Factor Range       | -0.8~+0.8             |

|                              |      |
|------------------------------|------|
| Max.Grid Passthrough Current | 200A |
|------------------------------|------|

**AC Output (EPS)**

|                         |                       |
|-------------------------|-----------------------|
| Rated Output Power      | 12kVA                 |
| Rated Output Current    | 50A                   |
| EPS Voltage             | 120V/240V Split phase |
| EPS Frequency(Optional) | 60Hz                  |

**Battery**

|                          |         |
|--------------------------|---------|
| Battery Voltage Range    | 40V~58V |
| Max. charging Current    | 250A    |
| Max. Discharging Current | 260A    |

**System**

|                    |                             |
|--------------------|-----------------------------|
| Protection Degree: | IP65/NEMA 3R                |
| Dimensions(W*D*H)  | 495*900*260mm               |
| Weight             | 56Kg                        |
| Max. Efficiency    | 96.5%                       |
| PV DC AFCI         | Type 1                      |
| PVRSS              | Tigo RSS transmitters (RS2) |

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


**MEGAREVO**  
Hybrid Solar Inverter R16KLNA-P6  
Grid Support Interactive Inverter

**PV Input**

|                           |          |
|---------------------------|----------|
| Max PV Input              | 24kW     |
| Mppt input Voltage        | 120~430V |
| MAX.Input Current         | 20A*4    |
| MAX.Short Circuit Current | 25A*4    |

**AC Output (On Grid)**

|                          |                       |
|--------------------------|-----------------------|
| Rated Output Power       | 16kVA                 |
| Rated Output current     | 66.7A                 |
| Grid Voltage             | 120V/240V Split phase |
| Grid Frequency(Optional) | 60Hz                  |
| Power Factor Range       | -0.8~+0.8             |

|                              |      |
|------------------------------|------|
| Max.Grid Passthrough Current | 200A |
|------------------------------|------|

**AC Output (EPS)**

|                         |                       |
|-------------------------|-----------------------|
| Rated Output Power      | 13kVA                 |
| Rated Output Current    | 54.2A                 |
| EPS Voltage             | 120V/240V Split phase |
| EPS Frequency(Optional) | 60Hz                  |

**Battery**

|                          |         |
|--------------------------|---------|
| Battery Voltage Range    | 40V~58V |
| Max. charging Current    | 260A    |
| Max. Discharging Current | 280A    |

**System**

|                    |                             |
|--------------------|-----------------------------|
| Protection Degree: | IP65/NEMA 3R                |
| Dimensions(W*D*H)  | 495*900*260mm               |
| Weight             | 56Kg                        |
| Max. Efficiency    | 96.5%                       |
| PV DC AFCI         | Type 1                      |
| PVRSS              | Tigo RSS transmitters (RS2) |

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

**Note:**

1. The above markings are the minimum requirements required by the safety standard. For the final production samples, the additional markings which do not give rise to misunderstanding may be added.
2. Label is attached on the side surface of enclosure and visible after installation
3. SGS NA Listed Mark shall be at least 5 mm high. Other letters shall be at least 2.4 mm high.
4. "CAUTION" shall be at lease 3.2mm high, the remaining letters shall not be less tan 1.6mm high
5. The S/N number should be pasted on every end product, every S/N number denotes the date code or other dating of manufacture.

Equipment under testing (\*):

- **R12KLNA-P6**

Variant Models:

- R8KLNA-P4
- R12KLNA-P4
- R16KLNA-P4
- R8KLNA-P6
- R16KLNA-P6

(\*) Fully tests according all points of the standards have been applied to model R12KLNA-P6.

The specific difference between R(8-16)KLNA-P4 and R(8-16)KLNA-P6 is that the wiring compartment of R(8-16)KLNA-P4 does not have a circuit breaker, while the wiring compartment of R(8-16)KLNA-P6 has input and output breakers.

Following table shows the full ratings of the all models referenced in this report, marked in **bold letters** the ones subjected to testing:

| Model:                                                                   | R8KLNA-P4                                                             | R12KLNA-P4           |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------|
| INPUT RATINGS:                                                           |                                                                       |                      |
| Maximum input voltage                                                    | 500 V dc                                                              |                      |
| Range of input operating voltage (without Battery)                       | 120 V dc to 500 V dc                                                  | 120 V dc to 500 V dc |
| Range of input operating voltage (with Battery)                          | 120V dc to 430 V dc                                                   | 120V dc to 430 V dc  |
| Maximum input current (dc)                                               | 16A/16A/16A/16A                                                       | 16A/16A/16A/16A      |
| Number of input                                                          | 4                                                                     |                      |
| OUTPUT RATINGS (Grid terminal, Bi-directional):                          |                                                                       |                      |
| Output power factor rating                                               | default >0.99 (-0.8~0.8 adjustable)                                   |                      |
| Operating voltage range (ac) (L-L)                                       | 0.88Un~1.1Un                                                          |                      |
| Operating frequency range or single frequency                            | 58.5 Hz to 61.2 Hz                                                    |                      |
| Number of phases                                                         | Single phase/ Split phase                                             |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac, split phase, 240V single phase                         |                      |
| Normal output frequency                                                  | 60 Hz                                                                 |                      |
| Maximum continuous output current (ac) per line                          | 36.7 Arms                                                             | 55.0 Arms            |
| Rated output current (ac) per line                                       | 33.3 Arms                                                             | 55.0 Arms            |
| Maximum continuous output power (ac)                                     | 8.8 kVA                                                               | 13.2 kVA             |
| Rated continuous output power (ac)                                       | 8.0 kW                                                                | 12.0 kW              |
| Maximum output fault current (ac) and duration                           | 378Apk, 83ms, 71.1 Arms@1cycle;<br>97.7 Arms@3cycle; 97.7 Arms@5cycle |                      |
| Trip limit and trip time accuracy - Voltage:                             | ±2.4 V                                                                | ±2.4 V               |
| Utility interconnection voltage and frequency trip limits and trip times | see Note 1 and 2                                                      |                      |
| Trip limit and trip time accuracy - Frequency:                           | ±0.01 Hz                                                              |                      |
| Trip limit and trip time accuracy - Time                                 | ±1%setting, but not less than 50ms                                    |                      |
| Normal operation temperature range                                       | -25°C to 60°C                                                         |                      |
| Enclosure Rating Type                                                    | Type 3R                                                               |                      |
| Weigh (kg)                                                               | 55kg                                                                  | 55kg                 |
| Dimension (mm)                                                           | 495*900*260                                                           |                      |
| OUTPUT RATINGS (BACKUP output terminal):                                 |                                                                       |                      |
| Number of phases                                                         | Single phase/Split phase                                              |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac                                                         |                      |
| Normal output frequency                                                  | 60 Hz                                                                 |                      |
| Maximum continuous output power (ac)                                     | 8.8 kVA                                                               | 13.2kVA              |

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|                                          |                   |              |
|------------------------------------------|-------------------|--------------|
| <b>Model:</b>                            | R8KLNA-P4         | R12KLNA-P4   |
| Rated output power (ac)                  | 8.0 kW            | 12.0 kW      |
| <b>Battery terminal, Bi-directional:</b> |                   |              |
| Battery Type                             | Lithium/Lead acid |              |
| Range of DC operating voltage (Vdc)      | 40-58V dc         |              |
| Nominal voltage (Vdc)                    | 48V dc            |              |
| Max. charging/ discharging current (Adc) | 170/170 Arms      | 250/260 Arms |
| Max. charging/ discharging power (W)     | 8000W             | 12000W       |

|                                                                          |                                                                       |                      |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------|
| Model:                                                                   | R16KLNA-P4                                                            | R8KLNA-P6            |
| INPUT RATINGS:                                                           |                                                                       |                      |
| Maximum input voltage                                                    | 500 V dc                                                              |                      |
| Range of input operating voltage (without Battery)                       | 120 V dc to 500 V dc                                                  | 120 V dc to 500 V dc |
| Range of input operating voltage (with Battery)                          | 120V dc to 430 V dc                                                   | 120V dc to 430 V dc  |
| Maximum input current (dc)                                               | 20A/20A/20A/20A                                                       | 16A/16A/16A/16A      |
| Number of input                                                          | 4                                                                     |                      |
| OUTPUT RATINGS:                                                          |                                                                       |                      |
| Output power factor rating                                               | default >0.99 (-0.8~0.8 adjustable)                                   |                      |
| Operating voltage range (ac) (L-L)                                       | 0.88Un~1.1Un                                                          |                      |
| Operating frequency range or single frequency                            | 58.5 Hz to 61.2 Hz                                                    |                      |
| Number of phases                                                         | Single phase/ Split phase                                             |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac                                                         |                      |
| Normal output frequency                                                  | 60 Hz                                                                 |                      |
| Maximum continuous output current (ac) per line                          | 73.3 Arms                                                             | 36.7 Arms            |
| Rated output current (ac) per line                                       | 66.7 Arms                                                             | 33.3 Arms            |
| Maximum continuous output power (ac)                                     | 17.6 kVA                                                              | 8.8 kVA              |
| Rated continuous output power (ac)                                       | 16.0 kW                                                               | 8.0 kW               |
| Maximum output fault current (ac) and duration                           | 378Apk, 83ms, 71.1 Arms@1cycle;<br>97.7 Arms@3cycle; 97.7 Arms@5cycle |                      |
| Trip limit and trip time accuracy - Voltage:                             | ±2.4 V                                                                | ±2.4 V               |
| Utility interconnection voltage and frequency trip limits and trip times | see Note 1 and 2                                                      |                      |
| Trip limit and trip time accuracy - Frequency:                           | ±0.01 Hz                                                              |                      |
| Trip limit and trip time accuracy - Time                                 | ±1%setting, but not less than 50ms                                    |                      |
| Normal operation temperature range                                       | -25°C to 60°C                                                         |                      |
| Enclosure Rating Type                                                    | Type 3R                                                               |                      |
| Weigh (kg)                                                               | 55kg                                                                  | 56kg                 |
| Dimension (mm)                                                           | 495*900*260                                                           |                      |
| OUTPUT RATINGS (BACKUP output terminal):                                 |                                                                       |                      |
| Number of phases                                                         | Single phase/Split phase                                              |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac                                                         |                      |
| Normal output frequency                                                  | 60 Hz                                                                 |                      |
| Maximum continuous output power (ac)                                     | 13.2 kVA                                                              | 8.8 kVA              |
| Rated output power (ac)                                                  | 13.0 kW                                                               | 8.0 kW               |
| Battery terminal, Bi-directional:                                        |                                                                       |                      |
| Battery Type                                                             | Lithium/Lead acid                                                     |                      |
| Range of DC operating voltage (Vdc)                                      | 40-58V dc                                                             |                      |
| Nominal voltage (Vdc)                                                    | 48V dc                                                                |                      |
| Max. charging/ discharging current (Adc)                                 | 260/280 Arms                                                          | 170/170 Arms         |
| Max. charging/ discharging power (W)                                     | 16000W                                                                | 8000W                |

|                                                                                   |                                   |               |
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|                                                                          |                                                                                    |                      |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------|
| Model:                                                                   | R12KLNA-P6                                                                         | R16KLNA-P6           |
| INPUT RATINGS:                                                           |                                                                                    |                      |
| Maximum input voltage                                                    | 500 V dc                                                                           |                      |
| Range of input operating voltage (without Battery)                       | 120 V dc to 500 V dc                                                               | 120 V dc to 500 V dc |
| Range of input operating voltage (with Battery)                          | 120V dc to 430 V dc                                                                | 120V dc to 430 V dc  |
| Maximum input current (dc)                                               | 16A/16A/16A/16A                                                                    | 20A/20A/20A/20A      |
| Number of input                                                          | 4                                                                                  |                      |
| OUTPUT RATINGS:                                                          |                                                                                    |                      |
| Output power factor rating                                               | default >0.99 (-0.8~0.8 adjustable)                                                |                      |
| Operating voltage range (ac) (L-L)                                       | 0.88Un~1.1Un                                                                       |                      |
| Operating frequency range or single frequency                            | 58.5 Hz to 61.2 Hz                                                                 |                      |
| Number of phases                                                         | Single phase/ Split phase                                                          |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac                                                                      |                      |
| Normal output frequency                                                  | 60 Hz                                                                              |                      |
| Maximum continuous output current (ac) per line                          | 55.0 Arms                                                                          | 73.3 Arms            |
| Rated output current (ac) per line                                       | 55.0 Arms                                                                          | 66.7 Arms            |
| Maximum continuous output power (ac)                                     | 13.2 kVA                                                                           | 17.6 kVA             |
| Rated continuous output power (ac)                                       | 12.0 kW                                                                            | 16.0 kW              |
| Maximum output fault current (ac) and duration                           | 378A <sub>pk</sub> , 83ms, 71.1 Arms@1cycle;<br>97.7 Arms@3cycle; 97.7 Arms@5cycle |                      |
| Trip limit and trip time accuracy - Voltage:                             | ±2.4 V                                                                             | ±2.4 V               |
| Utility interconnection voltage and frequency trip limits and trip times | see Note 1 and 2                                                                   |                      |
| Trip limit and trip time accuracy - Frequency:                           | ±0.01 Hz                                                                           |                      |
| Trip limit and trip time accuracy - Time                                 | ±1%setting, but not less than 50ms                                                 |                      |
| Normal operation temperature range                                       | -25°C to 60°C                                                                      |                      |
| Enclosure Rating Type                                                    | Type 3R                                                                            |                      |
| Weigh (kg)                                                               | 56kg                                                                               | 56kg                 |
| Dimension (mm)                                                           | 495*900*260                                                                        |                      |
| OUTPUT RATINGS (BACKUP output terminal):                                 |                                                                                    |                      |
| Number of phases                                                         | Single phase/Split phase                                                           |                      |
| Nominal output voltage (ac)                                              | 120Vac/240Vac                                                                      |                      |
| Normal output frequency                                                  | 60 Hz                                                                              |                      |
| Maximum continuous output power (ac)                                     | 13.2kVA                                                                            | 13.2 kVA             |
| Rated output power (ac)                                                  | 12.0 kW                                                                            | 13.0 kW              |
| Battery terminal, Bi-directional:                                        |                                                                                    |                      |
| Battery Type                                                             | Lithium/Lead acid                                                                  |                      |
| Range of DC operating voltage (Vdc)                                      | 40-58V dc                                                                          |                      |
| Nominal voltage (Vdc)                                                    | 48V dc                                                                             |                      |
| Max. charging/ discharging current (Adc)                                 | 250/260 Arms                                                                       | 260/280 Arms         |
| Max. charging/ discharging power (W)                                     | 12000W                                                                             | 16000W               |

The results obtained apply only to the particular sample tested that is the subject of the present test report. The most unfavorable result values of the verifications and tests performed are contained herein. Throughout this report a point (comma) is used as the decimal separator.

### 2.3 SGS Test Equipment List

|       | No. | Equipment Name                       | MARK/Model No.                     | Equipment No.               | Equipment calibration due date                       |
|-------|-----|--------------------------------------|------------------------------------|-----------------------------|------------------------------------------------------|
| BALUN | 1   | Oscilloscope                         | Tektronix/MS04054B                 | BZ-DGD-L064                 | 2024/01/23 to 2025/01/22                             |
|       | 2   | Differential probe                   | Tektronix / TPP0500                | BZ-DGD-L028-1               | 2024/01/23 to 2025/01/22                             |
|       | 3   | Temperature Chamber                  | OUKE/OK-TH0512C                    | BZ-DGB-L254                 | 2023/09/07 to 2024/09/06                             |
|       | 4   | Temperature recorder                 | 日置/HIOKI/LR8400-21                 | BZ-DGD-L038                 | 2023/09/08 to 2024/09/07                             |
|       | 5   | Temperature & Humidity meter         | CEM/DT-322                         | BZ-DGD-L270                 | 2023/09/13 to 2024/09/12                             |
|       | 6   | ESD Test System                      | SCHLODER/ SESD 30000               | --                          |                                                      |
|       | 7   | Anechoic Chamber                     | YIHENG ELECTRONIC/ 12.0m*7.0m*7.5m | --                          |                                                      |
|       | 8   | Signal Generator                     | Agilent/ N5181A                    | BZ-DGE-L001                 | 2024/01/30 to 2025/01/29                             |
|       | 9   | Power Amplifier                      | rflight/ NTWPA-00810200E           | BZ-DGE-L019                 | 2024/01/22 to 2025/01/21                             |
|       | 10  | Power Amplifier                      | rflight/ NTWPA-1060100E            | BZ-DGE-L020                 | 2024/01/22 to 2025/01/21                             |
|       | 11  | Power Meter                          | Agilent/ E4417A                    | BZ-DGE-L002                 | 2023/03/08 to 2024/03/07<br>2024/03/08 to 2025/03/07 |
|       | 12  | Feld Strength Meter                  | Narda/EP601                        | BZ-DGE-L050                 | 2024/02/05 to 2025/02/04                             |
|       | 13  | EFT Test System                      | HTEC/ HEFT 51                      | BZ-DGE-L033                 | 2024/01/09 to 2025/01/08                             |
|       | 14  | EFT coupling network                 | HTEC/ HPFS 303P/HV3P30             | BZ-DGE-L031/<br>BZ-DGE-L032 | 2024/01/12 to 2025/01/11                             |
|       | 15  | SURGE Generator (AC/DC Ports)        | HTEC/HCWG 70                       | BZ-DGE-L036                 | 2023/06/08 to 2024/06/07                             |
|       | 16  | SURGE coupling network (AC/DC Ports) | HTEC/SCDN303P7                     | BZ-DGE-L035                 | 2024/01/25 to 2025/01/24                             |
|       | 17  | Conducted immunity system            | Schlode/ CDG 6000-75               | BZ-DGE-L021                 | 2023/09/18 to 2024/09/17                             |
|       | 18  | Voltage withstand instrument         | Chroma/ 19032                      | BZ-DGD-L066                 | 2024/01/23 to 2025/01/22                             |

|     |    |                     |              |           |                          |
|-----|----|---------------------|--------------|-----------|--------------------------|
| SGS | 19 | True RMS Multimeter | FLUKE / 177C | GZE012-71 | 2023/11/06 to 2024/11/05 |
|-----|----|---------------------|--------------|-----------|--------------------------|

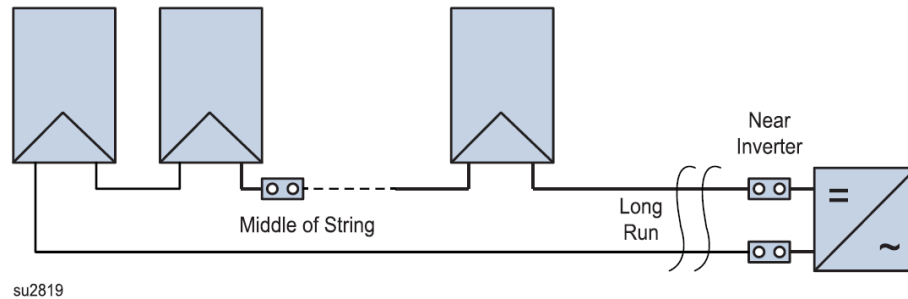
### 2.4 Measurement Uncertainty

Associated uncertainties through measurements showed in this this report are the maximum allowable uncertainties.

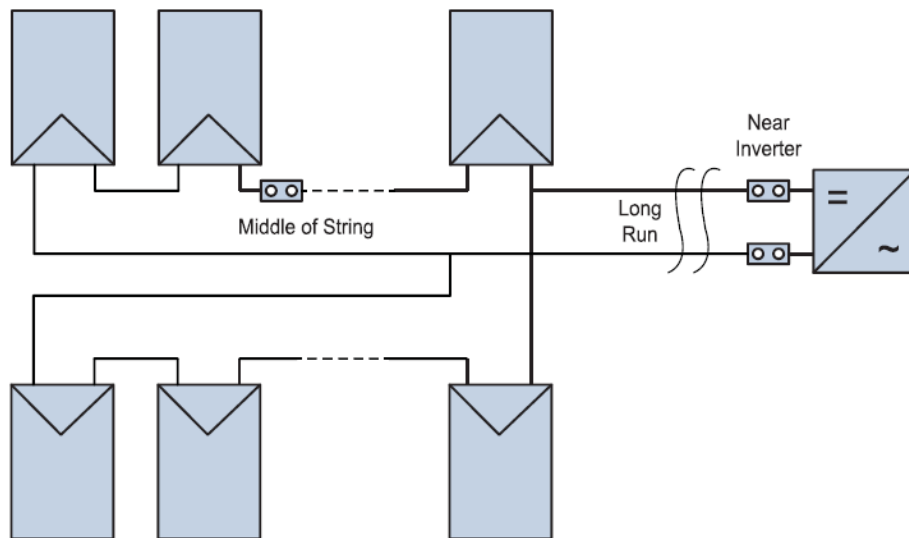
| Magnitude                                                                                                                                                                                                                                 | Uncertainty             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Voltage measurement                                                                                                                                                                                                                       | $\pm 0.05 \%$           |
| Current measurement                                                                                                                                                                                                                       | $\pm 0.05 \%$           |
| Frequency measurement                                                                                                                                                                                                                     | $\pm 0.001 \text{ Hz}$  |
| Time measurement                                                                                                                                                                                                                          | $\pm 0.001 \text{ s}$   |
| Power measurement                                                                                                                                                                                                                         | $\pm 0.5 \%$            |
| Phase Angle                                                                                                                                                                                                                               | $\pm 0.1^\circ$         |
| Temperature                                                                                                                                                                                                                               | $\pm 3^\circ \text{ C}$ |
| Note1: Measurements uncertainties showed in this table are maximum allowable uncertainties. The measurement uncertainties associated with other parameters measured during the tests are in the laboratory at disposal of the solicitant. |                         |

## 2.5 Test set up.

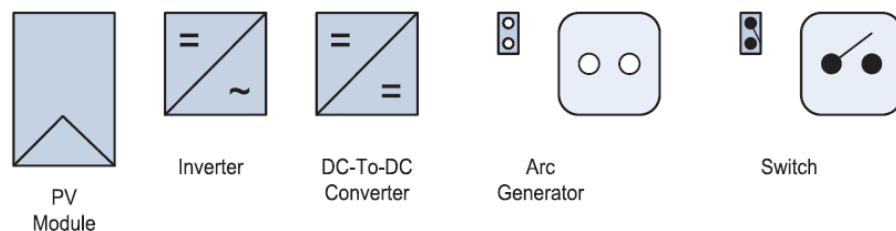
Below is the simplified construction of the test set up.



**One string, one MPPT**



**Two strings combined, one MPPT**



### Legend

Different equipment has been used to take measures as it shows in chapter 2.3. Current and voltage clamps have been connected to the inverter input / output for all the tests.

All the tests described in the following pages have used this specified test setup.

**The test bench used includes:**

| EQUIPMENT     | MARK / MODEL             | RATED CHARACTERISTICS                         | OWNER / ID.CODE |
|---------------|--------------------------|-----------------------------------------------|-----------------|
| AC source     | Chroma / WPLA-330 200KVA | 5-325Vac/ 5-650Vac/ 45-65Hz<br>0-320A/ 0-160A | --              |
| DC source     | WPLA-150KW               | 50 – 1500 Vdc;<br>0 – 200 A                   |                 |
| RLC load      | Qunlin / ACLT38160H      | 563.3kW, 563.3 kVAr                           | --              |
| Arc Generator | B.R. SOLAR               | --                                            | --              |

## 2.6 Definitions

|      |                                |     |                      |
|------|--------------------------------|-----|----------------------|
| In   | Nominal Current                | P   | Power                |
| p.u. | Per unit                       | I   | Current              |
| Pn   | Nominal Power                  | PV  | Photovoltaica        |
| Sn   | Apparent Power                 | f   | Frequency            |
| PGU  | Power Generation Unit          | W   | Watt                 |
| S/C  | Short Circuit                  | V   | Volt                 |
| O/C  | Open Circuit                   | A   | Ampere               |
| ms   | Millisecond                    | Un  | Nominal Voltage      |
| AFCI | Arc-Fault Circuit-Interrupters | EUT | Equipment under test |
| Hz   | Hertz                          |     |                      |

### 3 RESUME OF TEST RESULTS

#### INTERPRETATION KEYS

Test object does meet the requirement ..... **P** Pass  
 Test object does not meet the requirement ..... **F** Fails  
 Test case does not apply to the test object ..... **N/A** Not applicable  
 To make a reference to a table or an annex. .... See additional sheet  
 To indicate that the test has not been realized ..... **N/R** Not realized

| REPORT<br>CLAUSE | STANDARD<br>CLAUSE | STANDARD REQUIREMENTS                                                          |         | RESULTS |
|------------------|--------------------|--------------------------------------------------------------------------------|---------|---------|
|                  |                    | UL 1699B: Revision, Dated May 18, 2021                                         |         |         |
|                  |                    | TEST                                                                           | REMARKS |         |
| 4.1              | 25                 | Humidity conditioning                                                          |         | P       |
| 4.2              | 27                 | Voltage surge test                                                             |         | P       |
| 4.2.1            | 27.2               | Unwanted tripping test (ring wave)                                             |         | P       |
| 4.2.2            | 27.3               | Unwanted tripping test (ring wave)                                             |         | P       |
| 4.3.1            | 28                 | Environmental test sequence                                                    |         | P       |
| 4.3.2            | 29                 | Arc fault detection tests                                                      |         | P       |
| 4.4              | 30                 | Unwanted tripping tests                                                        |         | P       |
| 4.4.1            | 30.2               | Loading condition i – inverters, converters, and charge controllers            |         | P       |
| 4.4.2            | 30.3               | Loading condition ii – dc switch operation                                     |         | P       |
| 4.4.3            | 30.4               | Loading condition iii – irradiance step changes                                |         | P       |
| 4.5              | 31                 | Normal temperature test                                                        |         | P       |
| 4.6              | 35                 | Dielectric voltage-withstand test                                              |         | P       |
| 4.7              | 42                 | Resistance to environmental noise test                                         |         | P       |
| 4.8              | 55                 | Voltage surge test                                                             |         | P       |
| 4.9              | 56                 | Overvoltage, overload, endurance, abnormal operations, and short circuit tests |         | P       |

Note: Decision rule of the declaration of conformity evaluated according to the ILAC G8: 09/2019 & IEC 115 Guidelines (Proc. 2 “Accuracy Method” based on OD-5014).  
 Decision rule used: Binary with simple acceptance. (Safety Zone with respect to the limit  $w = 0$ ).  
 Specific risk: Probability of False Acceptance or Rejection less than 50%, (PFA / PFR < 50%). For more information see ILAC Guide G8 / 09.

## 4 TEST RESULTS

### 4.1 HUMIDITY CONDITIONING

The verification of Humidity Conditioning has been measured according to the chapter 25 of the standard.

A representative device is to be exposed for 168 hours to air at a relative humidity of  $93 \pm 2$  percent at a temperature of  $32.0 \pm 2.0^{\circ}\text{C}$  ( $89.6 \pm 3.6^{\circ}\text{F}$ ). The device is to be exposed to ambient air at a temperature of at least  $30^{\circ}\text{C}$  ( $86^{\circ}\text{F}$ ) until thermal equilibrium is attained before being placed in the test chamber.

Following the conditioning, while still in the chamber or within 60 s after removal from the chamber, the representative device shall be connected to an appropriate source of power. Operation of the representative device shall be verified by actuating the test circuit.

**Test result: PASS**

|                | Temperature( $^{\circ}\text{C}$ ) | Humidity(%) | Result             |
|----------------|-----------------------------------|-------------|--------------------|
| Test Condition | 32                                | 93          | EUT works normally |

## 4.2 VOLTAGE SURGE TEST

The verification of Humidity Conditioning has been measured according to the chapter 27 of the standard.

Devices deriving their power from a photovoltaic DC source shall be subjected to the tests described in 27.2 and 27.3 with the surges applied to the PV DC input of the device

### 4.2.1 UNWANTED TRIPPING TEST (RING WAVE)

All devices shall still be functional following the application of the ring wave surges. Functionality of a device shall be verified by connecting it to an appropriate power source and actuating the test circuit.

**Test result: PASS**

| Positive to Negative(1500V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|--------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                               | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                  | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                   | 870             | 870     | 870     | 870     | 870     | 870             | 870     | 880     | 870     | 870     |
| I (A)                                                  | 570             | 570     | 570     | 570     | 570     | 580             | 580     | 580     | 580     | 580     |

| Positive to Ground (3000V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|-------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                              | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                  | 2820            | 2830    | 2830    | 2830    | 2820    | 2810            | 2810    | 2810    | 2810    | 2810    |
| I (A)                                                 | 140             | 140     | 140     | 140     | 140     | 170             | 170     | 170     | 170     | 170     |

| Negative to Ground (3000V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|-------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                              | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                  | 2800            | 2800    | 2800    | 2800    | 2800    | 2830            | 2830    | 2830    | 2830    | 2830    |
| I (A)                                                 | 160             | 160     | 160     | 160     | 160     | 150             | 150     | 150     | 150     | 150     |

#### 4.2.2 SURGE IMMUNITY TEST (COMBINATION WAVE)

Following the application of the Surge Immunity (Combination Wave) surges, the device shall be in a condition to continue the test sequence in Table 24.1. Functionality of the device shall be verified by connecting it to an appropriate power source and actuating the supervisory circuit.

**Table 27.1**  
**PV input surge requirements**

| Surge type                             | Surge/impulse voltage | Surge test resistance | Number of impulses per polarity | Surge polarity for each test point |
|----------------------------------------|-----------------------|-----------------------|---------------------------------|------------------------------------|
| Combination wave<br>(1.2/50 $\mu$ sec) | Table 27.2            | 12 ohms               | 10                              | Negative                           |
|                                        |                       |                       | 10                              | Positive                           |

**Table 27.2**  
**Surge values**

| Maximum dc input rating of equipment | Test Surge/impulse voltage |
|--------------------------------------|----------------------------|
| 71 Vdc                               | 500 V                      |
| 141 Vdc                              | 800 V                      |
| 213 Vdc                              | 1500 V                     |
| 424 Vdc                              | 2500 V                     |
| 849 Vdc                              | 4000 V                     |
| 1500 Vdc                             | 6000 V                     |
| Note: Interpolation is permitted     |                            |

**Test result: PASS**

| Positive to Negative(1400V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|--------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                               | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                  | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                   | 810             | 810     | 810     | 810     | 810     | 810             | 810     | 810     | 810     | 810     |
| I (A)                                                  | 520             | 520     | 520     | 520     | 520     | 540             | 540     | 540     | 540     | 540     |

| Positive to Ground (2800V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|-------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                              | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                  | 2630            | 2630    | 2640    | 2630    | 2630    | 2620            | 2620    | 2620    | 2620    | 2620    |
| I (A)                                                 | 130             | 130     | 130     | 130     | 130     | 150             | 150     | 150     | 150     | 150     |

| Negative to Ground (2800V): 60 seconds between pulses |                 |         |         |         |         |                 |         |         |         |         |
|-------------------------------------------------------|-----------------|---------|---------|---------|---------|-----------------|---------|---------|---------|---------|
| Polarity                                              | Positive Pulses |         |         |         |         | Negative Pulses |         |         |         |         |
| Pulse                                                 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 | Pulse 1         | Pulse 2 | Pulse 3 | Pulse 4 | Pulse 5 |
| U(V)                                                  | 2610            | 2610    | 2610    | 2610    | 2610    | 2640            | 2640    | 2640    | 2640    | 2640    |
| I (A)                                                 | 150             | 150     | 150     | 150     | 150     | 130             | 130     | 130     | 130     | 130     |

### 4.3 ENVIRONMENTAL TEST SEQUENCE AND ARC FAULT DETECTION TESTS

#### 4.3.1 ENVIRONMENTAL TEST SEQUENCE

The verification of Environmental Test Sequence has been measured according to the chapter 28 of the standard.

A device that has been conditioned by humidity, leakage current (when required), and voltage surge, shall comply with all of the applicable tests in Arc Fault Detection Tests, Section 29, while operating in ambient air at 25°C (77°F). The same representative device shall also respond to the presence of an arcing fault when subjected to a repeated Series Connection Arc Test, while in ambient air at 65°C (149°F), -35°C (-31°F), and 25°C (77°F) by following the sequence shown in Table 28.1.

**Table 28.1**  
**Environmental test sequence for arc fault detection tests**

| Ambient air temperature <sup>a</sup>          | Operating parameters      | Remarks                                                                         |
|-----------------------------------------------|---------------------------|---------------------------------------------------------------------------------|
| 1. 25.0 ±5.0°C<br>(77.0 ±9.0°F)               | No voltage applied        | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 2. 25.0 ±2.0°C<br>(77.0 ±3.6°F)               | Rated voltage             | Test per 29.1 as soon as possible to minimize self-heating.                     |
| 3. <sup>b</sup> 65.0 ±2.0°C<br>(149.0 ±3.6°F) | Rated voltage and current | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 4. <sup>b</sup> 65.0 ±2.0°C<br>(149.0 ±3.6°F) | Rated voltage             | Test per 29.1.                                                                  |
| 5. <sup>c</sup> 40.0 ±2.0°C<br>(104.0 ±3.6°F) | Rated voltage and current | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 6. <sup>c</sup> 40.0 ±2.0°C<br>(104.0 ±3.6°F) | Rated voltage             | Test per 29.1.                                                                  |
| 7. 25.0 ±5.0°C<br>(77.0 ±9.0°F)               | No voltage applied        | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 8. <sup>d</sup> -35.0 ±2.0°C<br>(-31 ±3.6°F)  | No voltage applied        | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 9. <sup>d</sup> -35.0 ±2.0°C<br>(-31 ±3.6°F)  | Rated voltage             | Tests per 29.1 as soon as possible to minimize self-heating                     |
| 10. 25.0 ±5.0°C<br>(77.0 ±9.0°F)              | Rated voltage and current | Establish thermal equilibrium with at least two hours of exposure. Do not test. |
| 11. 25.0 ±5.0°C<br>(77.0 ±9.0°F)              | Rated voltage             | Test per 29.1.                                                                  |

<sup>a</sup> The ambient air temperature is to be changed to each value shown without intentional delay.

<sup>b</sup> For devices specified for use in higher ambient air conditions, such as devices for mounting directly to some modules or some rooftops, the device shall be tested at the maximum specified temperature extreme. In the event that a device is self-protecting such that it trips or functions at this ambient temperature, lower values of load current are to be employed, until the device just continues to operate, if possible.

<sup>c</sup> This test is not to be performed if steps 3 and 4 have been performed employing rated current.

<sup>d</sup> For devices specified for use in lower ambient air conditions, the device shall be tested at the minimum specified temperature extreme.

#### 4.3.2 ARC FAULT DETECTION TESTS

The verification of Arc Fault Detection Tests has been measured according to the chapter 28, 29.1, 29.2 and 29.3 of the standard.

Arcing test conditions:

**Table 29.2**  
**Arcing test conditions**

| Test no.                                                                                                                                                                                                                                                                                                                                                               | Minimum $I_{arc}$ (A) <sup>a</sup> | $I_{mpp}$ (A) | Sep. rate (mm/s) | $V_{mpp}$ (V) <sup>b</sup> | $V_{oc}$ (V) <sup>b</sup> | $R_{tot}$ (ohms) <sup>b</sup> | Gap (mm) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------|------------------|----------------------------|---------------------------|-------------------------------|----------|
| 1                                                                                                                                                                                                                                                                                                                                                                      | 2.5                                | 3.0           | 2.5              | 312.0                      | 480.0                     | 56.0                          | 0.8      |
| 2                                                                                                                                                                                                                                                                                                                                                                      | 7.0                                | 8.0           | 5.0              | 318.0                      | 490.0                     | 21.0                          | 0.8      |
| 3 <sup>c</sup>                                                                                                                                                                                                                                                                                                                                                         | 14.0                               | 16.0          | 5.0              | 318.0                      | 490.0                     | 11.0                          | 1.1      |
| 4                                                                                                                                                                                                                                                                                                                                                                      | 7.0                                | 8.5           | 5.0              | 607.0                      | 810.0                     | 24.0                          | 2.5      |
| <b>Single module</b>                                                                                                                                                                                                                                                                                                                                                   |                                    |               |                  |                            |                           |                               |          |
| 5                                                                                                                                                                                                                                                                                                                                                                      | 2.5                                | 3.0           | 2.5              | 31.2                       | 48.0                      | 5.6                           | 0.8      |
| 6                                                                                                                                                                                                                                                                                                                                                                      | 7.0                                | 8.0           | 5.0              | 31.8                       | 49.0                      | 2.1                           | 0.8      |
| <sup>a</sup> $I_{arc}$ values in Table 29.2 are representative of realistic arc events with one or two strings at full and low irradiance. It is expected that the AFCI shall meet compliance criteria below at intermediate current levels also.<br><sup>b</sup> These values are approximate.<br><sup>c</sup> Required for arcs in series with two parallel strings. |                                    |               |                  |                            |                           |                               |          |

**Test Result: PASS**, one string, one MPPT

| Tem. (C) | Test No. | Location             | Attempt | $V_{arc}$ (V) | $I_{arc}$ (A) | $P_{arc}$ (W) | Trip Time measured(s) | Trip Time limited(s) | Pass /Fail |
|----------|----------|----------------------|---------|---------------|---------------|---------------|-----------------------|----------------------|------------|
| 25°C     | 1        | A (Start of String)  | 1       | 22.2          | 7.8           | 151.9         | 1.682                 | 2.5                  | Pass       |
|          |          |                      | 2       | 18.8          | 7.6           | 102.6         | 1.517                 | 2.5                  | Pass       |
|          |          |                      | 3       | 23.4          | 7.4           | 128.8         | 1.145                 | 2.5                  | Pass       |
|          |          |                      | 4       | 22.3          | 7.6           | 132.9         | 1.300                 | 2.5                  | Pass       |
|          |          |                      | 5       | 22.2          | 7.8           | 146.7         | 1.754                 | 2.5                  | Pass       |
|          |          | B (End of String)    | 1       | 17.2          | 7.5           | 94.2          | 1.230                 | 2.5                  | Pass       |
|          |          |                      | 2       | 21.0          | 7.9           | 148.3         | 1.496                 | 2.5                  | Pass       |
|          |          |                      | 3       | 23.4          | 7.6           | 149.3         | 1.631                 | 2.5                  | Pass       |
|          |          |                      | 4       | 13.0          | 7.7           | 72.4          | 1.465                 | 2.5                  | Pass       |
|          |          |                      | 5       | 19.1          | 7.6           | 110.4         | 1.578                 | 2.5                  | Pass       |
|          |          | C (Middle of String) | 1       | 145.0         | 3.0           | 56.5          | 1.218                 | 2.5                  | Pass       |
|          |          |                      | 2       | 142.2         | 2.7           | 10.4          | 0.831                 | 2.5                  | Pass       |
|          |          |                      | 3       | 124.6         | 2.8           | 45.4          | 0.717                 | 2.5                  | Pass       |
|          |          |                      | 4       | 127.2         | 2.7           | 4.6           | 0.831                 | 2.5                  | Pass       |
|          |          |                      | 5       | 178.7         | 2.8           | 21.2          | 0.437                 | 2.5                  | Pass       |
|          | 2        | A (Start of String)  | 1       | 18.2          | 7.2           | 105.3         | 2.425                 | 2.5                  | Pass       |
|          |          |                      | 2       | 22.2          | 7.4           | 136.8         | 2.187                 | 2.5                  | Pass       |
|          |          |                      | 3       | 21.9          | 7.6           | 145.9         | 2.053                 | 2.5                  | Pass       |
|          |          |                      | 4       | 23.2          | 7.4           | 134.7         | 1.135                 | 2.5                  | Pass       |
|          |          |                      | 5       | 17.1          | 7.5           | 89.1          | 1.568                 | 2.5                  | Pass       |
|          |          | B (End of String)    | 1       | 25.9          | 7.1           | 144.1         | 1.212                 | 2.5                  | Pass       |
|          |          |                      | 2       | 17.4          | 7.3           | 78.8          | 1.438                 | 2.5                  | Pass       |
|          |          |                      | 3       | 20.2          | 7.5           | 127.6         | 1.867                 | 2.5                  | Pass       |

|  |   |                         |   |       |      |       |       |     |      |
|--|---|-------------------------|---|-------|------|-------|-------|-----|------|
|  | 3 | C<br>(Middle of String) | 4 | 14.8  | 7.4  | 103.3 | 1.475 | 2.5 | Pass |
|  |   |                         | 5 | 2.9   | 7.5  | 12.5  | 2.053 | 2.5 | Pass |
|  |   |                         | 1 | 21.6  | 7.6  | 129.7 | 1.589 | 2.5 | Pass |
|  |   |                         | 2 | 10.7  | 7.2  | 73.2  | 1.331 | 2.5 | Pass |
|  |   |                         | 3 | 22.7  | 7.7  | 149.4 | 1.898 | 2.5 | Pass |
|  |   | A<br>(Start of String)  | 4 | 22.7  | 7.7  | 149.5 | 1.950 | 2.5 | Pass |
|  |   |                         | 5 | 24.1  | 7.2  | 128.8 | 0.991 | 2.5 | Pass |
|  |   |                         | 1 | 86.8  | 14.9 | 99.4  | 1.857 | 2.5 | Pass |
|  |   |                         | 2 | 21.1  | 14.5 | 244.1 | 1.682 | 2.5 | Pass |
|  |   |                         | 3 | 20.5  | 14.5 | 237.7 | 1.775 | 2.5 | Pass |
|  |   |                         | 4 | 21.6  | 15.3 | 283.8 | 1.496 | 2.5 | Pass |
|  |   |                         | 5 | 22.3  | 15.3 | 293.1 | 1.465 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 138.1 | 15.1 | 220.8 | 1.026 | 2.5 | Pass |
|  |   |                         | 2 | 23.9  | 14.4 | 272.8 | 1.228 | 2.5 | Pass |
|  |   |                         | 3 | 21.5  | 14.8 | 249.6 | 1.042 | 2.5 | Pass |
|  |   |                         | 4 | 18.8  | 14.0 | 186.1 | 1.867 | 2.5 | Pass |
|  |   |                         | 5 | 19.7  | 14.0 | 220.5 | 2.094 | 2.5 | Pass |
|  | 4 | C<br>(Middle of String) | 1 | 22.2  | 15.2 | 265.6 | 1.671 | 2.5 | Pass |
|  |   |                         | 2 | 11.1  | 14.9 | 159.5 | 1.661 | 2.5 | Pass |
|  |   |                         | 3 | 16.2  | 14.3 | 119.0 | 1.651 | 2.5 | Pass |
|  |   |                         | 4 | 20.6  | 14.3 | 243.5 | 1.929 | 2.5 | Pass |
|  |   |                         | 5 | 21.1  | 14.0 | 245.7 | 1.785 | 2.5 | Pass |
|  |   | A<br>(Start of String)  | 1 | 22.1  | 7.8  | 151.9 | 1.682 | 2.5 | Pass |
|  |   |                         | 2 | 18.8  | 7.6  | 102.5 | 1.517 | 2.5 | Pass |
|  |   |                         | 3 | 23.4  | 7.4  | 128.8 | 1.145 | 2.5 | Pass |
|  |   |                         | 4 | 22.3  | 7.6  | 132.9 | 1.300 | 2.5 | Pass |
|  |   |                         | 5 | 22.1  | 7.7  | 146.7 | 1.754 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 17.2  | 7.5  | 94.2  | 1.230 | 2.5 | Pass |
|  |   |                         | 2 | 21.0  | 7.9  | 148.3 | 1.496 | 2.5 | Pass |
|  |   |                         | 3 | 23.4  | 7.6  | 149.3 | 1.630 | 2.5 | Pass |
|  |   |                         | 4 | 13.0  | 7.7  | 72.3  | 1.465 | 2.5 | Pass |
|  |   |                         | 5 | 19.0  | 7.5  | 110.4 | 1.578 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 22.6  | 7.5  | 126.4 | 1.341 | 2.5 | Pass |
|  |   |                         | 2 | 20.2  | 7.7  | 122.5 | 1.867 | 2.5 | Pass |
|  |   |                         | 3 | 20.8  | 7.7  | 125.7 | 1.816 | 2.5 | Pass |
|  |   |                         | 4 | 19.0  | 7.9  | 134.4 | 1.661 | 2.5 | Pass |
|  |   |                         | 5 | 20.9  | 7.5  | 121.4 | 1.166 | 2.5 | Pass |

Note:

- After evaluation, when C4=150nF, the test conditions are the worst.

| Tem.<br>(C) | Test<br>No. | Location               | Attempt | Varc (V) | Iarc (A) | Parc (W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pass<br>/Fail |
|-------------|-------------|------------------------|---------|----------|----------|----------|--------------------------|-------------------------|---------------|
| -25°C       | 1           | A<br>(Start of String) | 1       | 95.1     | 2.5      | 38.8     | 0.511                    | 2.5                     | Pass          |
|             |             |                        | 2       | 87.7     | 2.6      | 85.4     | 1.898                    | 2.5                     | Pass          |
|             |             |                        | 3       | 86.9     | 2.5      | 45.1     | 0.923                    | 2.5                     | Pass          |
|             |             |                        | 4       | 124.1    | 2.6      | 20.3     | 1.135                    | 2.5                     | Pass          |
|             |             |                        | 5       | 83.9     | 2.5      | 49.0     | 0.768                    | 2.5                     | Pass          |
|             |             | B                      | 1       | 188.1    | 2.7      | 53.6     | 0.830                    | 2.5                     | Pass          |

|  |   |                         |   |       |      |       |       |     |      |
|--|---|-------------------------|---|-------|------|-------|-------|-----|------|
|  | 2 | (End of String)         | 2 | 53.0  | 2.9  | 94.7  | 1.558 | 2.5 | Pass |
|  |   |                         | 3 | 51.4  | 2.9  | 91.9  | 1.568 | 2.5 | Pass |
|  |   |                         | 4 | 70.2  | 2.5  | 53.6  | 0.598 | 2.5 | Pass |
|  |   |                         | 5 | 85.2  | 2.6  | 40.1  | 0.505 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 21.8  | 2.6  | 48.3  | 0.305 | 2.5 | Pass |
|  |   |                         | 2 | 101.3 | 2.8  | 92.6  | 1.754 | 2.5 | Pass |
|  |   |                         | 3 | 104.1 | 2.8  | 95.2  | 1.723 | 2.5 | Pass |
|  |   |                         | 4 | 45.8  | 2.9  | 79.7  | 1.578 | 2.5 | Pass |
|  |   |                         | 5 | 47.2  | 2.9  | 82.1  | 1.496 | 2.5 | Pass |
|  |   | A<br>(Start of String)  | 1 | 18.1  | 7.5  | 100.3 | 2.146 | 2.5 | Pass |
|  |   |                         | 2 | 20.6  | 7.9  | 143.0 | 2.156 | 2.5 | Pass |
|  |   |                         | 3 | 22.4  | 7.9  | 160.6 | 1.816 | 2.5 | Pass |
|  |   |                         | 4 | 23.0  | 7.7  | 147.8 | 1.847 | 2.5 | Pass |
|  |   |                         | 5 | 23.4  | 7.8  | 161.4 | 2.167 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 20.1  | 7.7  | 128.2 | 1.888 | 2.5 | Pass |
|  |   |                         | 2 | 22.7  | 7.6  | 141.3 | 1.971 | 2.5 | Pass |
|  |   |                         | 3 | 19.4  | 7.7  | 127.2 | 1.671 | 2.5 | Pass |
|  |   |                         | 4 | 20.2  | 7.3  | 120.2 | 1.981 | 2.5 | Pass |
|  |   |                         | 5 | 22.3  | 7.6  | 133.6 | 1.455 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 19.6  | 7.2  | 109.7 | 1.331 | 2.5 | Pass |
|  |   |                         | 2 | 19.3  | 7.4  | 111.6 | 1.960 | 2.5 | Pass |
|  |   |                         | 3 | 17.9  | 7.6  | 117.2 | 1.929 | 2.5 | Pass |
|  |   |                         | 4 | 19.7  | 7.7  | 132.5 | 2.156 | 2.5 | Pass |
|  |   |                         | 5 | 22.3  | 7.2  | 126.6 | 1.011 | 2.5 | Pass |
|  | 3 | A<br>(Start of String)  | 1 | 21.2  | 15.1 | 281.1 | 1.867 | 2.5 | Pass |
|  |   |                         | 2 | 21.3  | 14.6 | 237.1 | 1.651 | 2.5 | Pass |
|  |   |                         | 3 | 23.5  | 15.0 | 318.2 | 1.723 | 2.5 | Pass |
|  |   |                         | 4 | 22.9  | 15.1 | 309.8 | 1.754 | 2.5 | Pass |
|  |   |                         | 5 | 19.7  | 14.5 | 216.8 | 1.640 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 23.3  | 14.5 | 281.3 | 1.641 | 2.5 | Pass |
|  |   |                         | 2 | 22.9  | 15.5 | 309.7 | 1.898 | 2.5 | Pass |
|  |   |                         | 3 | 23.5  | 15.5 | 317.7 | 1.847 | 2.5 | Pass |
|  |   |                         | 4 | 21.7  | 14.1 | 247.6 | 1.744 | 2.5 | Pass |
|  |   |                         | 5 | 19.1  | 14.6 | 210.9 | 1.651 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 22.9  | 14.9 | 287.5 | 2.033 | 2.5 | Pass |
|  |   |                         | 2 | 22.4  | 14.9 | 280.9 | 2.063 | 2.5 | Pass |
|  |   |                         | 3 | 19.2  | 14.1 | 222.8 | 1.958 | 2.5 | Pass |
|  |   |                         | 4 | 18.8  | 14.2 | 217.5 | 1.981 | 2.5 | Pass |
|  |   |                         | 5 | 22.3  | 14.6 | 269.5 | 1.878 | 2.5 | Pass |
|  | 4 | A<br>(Start of String)  | 1 | 20.1  | 7.8  | 133.1 | 1.589 | 2.5 | Pass |
|  |   |                         | 2 | 21.5  | 7.9  | 143.3 | 1.785 | 2.5 | Pass |
|  |   |                         | 3 | 21.5  | 7.9  | 143.3 | 1.836 | 2.5 | Pass |
|  |   |                         | 4 | 20.4  | 7.3  | 114.1 | 1.888 | 2.5 | Pass |
|  |   |                         | 5 | 20.9  | 7.3  | 117.1 | 1.795 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 21.5  | 7.8  | 142.6 | 1.754 | 2.5 | Pass |
|  |   |                         | 2 | 19.8  | 7.8  | 128.2 | 1.640 | 2.5 | Pass |
|  |   |                         | 3 | 22.9  | 7.2  | 119.9 | 0.939 | 2.5 | Pass |
|  |   |                         | 4 | 19.7  | 7.9  | 136.4 | 1.971 | 2.5 | Pass |

|  |  |                         |   |      |     |       |       |     |      |
|--|--|-------------------------|---|------|-----|-------|-------|-----|------|
|  |  |                         | 5 | 20.2 | 7.9 | 139.8 | 1.929 | 2.5 | Pass |
|  |  | C<br>(Middle of String) | 1 | 12.2 | 7.2 | 72.0  | 1.372 | 2.5 | Pass |
|  |  |                         | 2 | 24.9 | 7.3 | 131.8 | 1.124 | 2.5 | Pass |
|  |  |                         | 3 | 23.5 | 7.3 | 120.5 | 0.825 | 2.5 | Pass |
|  |  |                         | 4 | 23.5 | 7.1 | 122.8 | 0.825 | 2.5 | Pass |
|  |  |                         | 5 | 18.3 | 7.4 | 104.3 | 1.898 | 2.5 | Pass |

Note:

- After evaluation, when C4=150nF, the test conditions are the worst.

| Tem.<br>(C) | Test<br>No. | Location                | Attempt | Varc (V) | Iarc (A) | Parc (W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pass<br>/Fail |
|-------------|-------------|-------------------------|---------|----------|----------|----------|--------------------------|-------------------------|---------------|
| 60°C        | 1           | A<br>(Start of String)  | 1       | 184.4    | 2.6      | 78.7     | 0.923                    | 2.5                     | Pass          |
|             |             |                         | 2       | 142.6    | 2.6      | 19.3     | 0.727                    | 2.5                     | Pass          |
|             |             |                         | 3       | 122.6    | 2.5      | 47.4     | 0.737                    | 2.5                     | Pass          |
|             |             |                         | 4       | 113.6    | 2.6      | 78.5     | 1.238                    | 2.5                     | Pass          |
|             |             |                         | 5       | 139.0    | 2.5      | 45.2     | 0.416                    | 2.5                     | Pass          |
|             |             | B<br>(End of String)    | 1       | 53.8     | 2.5      | 52.6     | 0.820                    | 2.5                     | Pass          |
|             |             |                         | 2       | 35.2     | 2.5      | 67.5     | 0.851                    | 2.5                     | Pass          |
|             |             |                         | 3       | 91.4     | 2.5      | 58.4     | 1.031                    | 2.5                     | Pass          |
|             |             |                         | 4       | 106.5    | 2.6      | 25.7     | 1.419                    | 2.5                     | Pass          |
|             |             |                         | 5       | 128.9    | 2.6      | 27.8     | 1.130                    | 2.5                     | Pass          |
|             |             | C<br>(Middle of String) | 1       | 124.4    | 2.5      | 37.4     | 0.520                    | 2.5                     | Pass          |
|             |             |                         | 2       | 94.0     | 2.5      | 71.1     | 0.954                    | 2.5                     | Pass          |
|             |             |                         | 3       | 117.9    | 2.5      | 35.2     | 0.634                    | 2.5                     | Pass          |
|             |             |                         | 4       | 174.9    | 2.5      | 62.7     | 0.841                    | 2.5                     | Pass          |
|             |             |                         | 5       | 133.1    | 2.8      | 88.9     | 1.362                    | 2.5                     | Pass          |
|             | 2           | A<br>(Start of String)  | 1       | 21.5     | 7.8      | 150.0    | 2.053                    | 2.5                     | Pass          |
|             |             |                         | 2       | 21.6     | 7.8      | 150.0    | 1.991                    | 2.5                     | Pass          |
|             |             |                         | 3       | 22.2     | 7.7      | 149.9    | 1.857                    | 2.5                     | Pass          |
|             |             |                         | 4       | 20.8     | 7.8      | 149.9    | 2.063                    | 2.5                     | Pass          |
|             |             |                         | 5       | 22.6     | 7.6      | 149.5    | 1.744                    | 2.5                     | Pass          |
|             |             | B<br>(End of String)    | 1       | 23.2     | 7.5      | 152.7    | 1.651                    | 2.5                     | Pass          |
|             |             |                         | 2       | 22.6     | 7.8      | 163.5    | 1.919                    | 2.5                     | Pass          |
|             |             |                         | 3       | 14.0     | 7.3      | 57.3     | 1.744                    | 2.5                     | Pass          |
|             |             |                         | 4       | 21.0     | 7.8      | 143.5    | 1.836                    | 2.5                     | Pass          |
|             |             |                         | 5       | 22.5     | 7.7      | 155.4    | 2.094                    | 2.5                     | Pass          |
|             |             | C<br>(Middle of String) | 1       | 18.4     | 7.9      | 132.8    | 2.053                    | 2.5                     | Pass          |
|             |             |                         | 2       | 20.1     | 7.8      | 142.4    | 2.156                    | 2.5                     | Pass          |
|             |             |                         | 3       | 22.2     | 7.9      | 158.0    | 1.878                    | 2.5                     | Pass          |
|             |             |                         | 4       | 23.1     | 7.1      | 126.9    | 0.949                    | 2.5                     | Pass          |
|             |             |                         | 5       | 24.4     | 7.5      | 168.2    | 2.043                    | 2.5                     | Pass          |
|             | 3           | A<br>(Start of String)  | 1       | 22.9     | 14.5     | 285.3    | 1.857                    | 2.5                     | Pass          |
|             |             |                         | 2       | 24.7     | 15.5     | 313.1    | 1.661                    | 2.5                     | Pass          |
|             |             |                         | 3       | 23.7     | 14.4     | 223.1    | 2.053                    | 2.5                     | Pass          |
|             |             |                         | 4       | 21.9     | 14.6     | 284.8    | 2.043                    | 2.5                     | Pass          |
|             |             |                         | 5       | 21.5     | 14.6     | 264.0    | 1.331                    | 2.5                     | Pass          |
|             |             | B<br>(End of            | 1       | 19.4     | 14.3     | 225.3    | 1.940                    | 2.5                     | Pass          |
|             |             |                         | 2       | 19.9     | 14.2     | 231.0    | 1.836                    | 2.5                     | Pass          |

|  |   |                         |   |      |      |       |       |     |      |
|--|---|-------------------------|---|------|------|-------|-------|-----|------|
|  | 4 | String)                 | 3 | 20.7 | 14.1 | 244.5 | 1.434 | 2.5 | Pass |
|  |   |                         | 4 | 99.7 | 14.2 | 110.9 | 1.682 | 2.5 | Pass |
|  |   |                         | 5 | 20.1 | 14.1 | 211.9 | 1.960 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 23.6 | 14.3 | 307.0 | 2.033 | 2.5 | Pass |
|  |   |                         | 2 | 23.1 | 14.4 | 300.0 | 2.063 | 2.5 | Pass |
|  |   |                         | 3 | 23.3 | 14.9 | 287.0 | 1.444 | 2.5 | Pass |
|  |   |                         | 4 | 22.6 | 14.9 | 277.9 | 1.455 | 2.5 | Pass |
|  |   |                         | 5 | 21.3 | 14.7 | 266.2 | 1.847 | 2.5 | Pass |
|  |   | A<br>(Start of String)  | 1 | 24.3 | 7.9  | 169.7 | 2.167 | 2.5 | Pass |
|  |   |                         | 2 | 18.5 | 7.2  | 85.1  | 1.166 | 2.5 | Pass |
|  |   |                         | 3 | 22.5 | 7.3  | 136.5 | 1.248 | 2.5 | Pass |
|  |   |                         | 4 | 17.6 | 7.4  | 60.8  | 1.207 | 2.5 | Pass |
|  |   |                         | 5 | 16.1 | 7.5  | 67.7  | 1.847 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 22.9 | 7.3  | 128.4 | 1.135 | 2.5 | Pass |
|  |   |                         | 2 | 21.7 | 7.9  | 145.1 | 1.651 | 2.5 | Pass |
|  |   |                         | 3 | 20.1 | 7.5  | 111.0 | 0.815 | 2.5 | Pass |
|  |   |                         | 4 | 16.7 | 7.9  | 102.6 | 1.867 | 2.5 | Pass |
|  |   |                         | 5 | 19.5 | 7.5  | 110.8 | 2.146 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 24.7 | 7.4  | 145.5 | 1.444 | 2.5 | Pass |
|  |   |                         | 2 | 20.0 | 7.8  | 135.5 | 1.661 | 2.5 | Pass |
|  |   |                         | 3 | 20.5 | 7.8  | 141.8 | 1.547 | 2.5 | Pass |
|  |   |                         | 4 | 21.9 | 7.5  | 129.9 | 1.341 | 2.5 | Pass |
|  |   |                         | 5 | 20.8 | 7.3  | 116.8 | 1.547 | 2.5 | Pass |

Note:

1. After evaluation, when C4=150nF, the test conditions are the worst.

| Tem.<br>(C) | Test<br>No. | Location                | Attempt | Varc (V) | Iarc (A) | Parc (W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pass<br>/Fail |
|-------------|-------------|-------------------------|---------|----------|----------|----------|--------------------------|-------------------------|---------------|
| 25°C        | 1           | A<br>(Start of String)  | 1       | 137.6    | 2.8      | 52.4     | 0.799                    | 2.5                     | Pass          |
|             |             |                         | 2       | 129.0    | 2.6      | 8.8      | 0.830                    | 2.5                     | Pass          |
|             |             |                         | 3       | 153.4    | 2.5      | 26.9     | 0.534                    | 2.5                     | Pass          |
|             |             |                         | 4       | 139.8    | 2.5      | 35.1     | 0.737                    | 2.5                     | Pass          |
|             |             |                         | 5       | 123.9    | 2.7      | 97.4     | 1.238                    | 2.5                     | Pass          |
|             |             | B<br>(End of String)    | 1       | 40.7     | 2.6      | 42.2     | 0.510                    | 2.5                     | Pass          |
|             |             |                         | 2       | 67.5     | 2.6      | 47.1     | 0.410                    | 2.5                     | Pass          |
|             |             |                         | 3       | 115.4    | 2.6      | 9.1      | 1.228                    | 2.5                     | Pass          |
|             |             |                         | 4       | 136.4    | 2.7      | 112.2    | 1.424                    | 2.5                     | Pass          |
|             |             |                         | 5       | 80.8     | 2.7      | 38.4     | 0.420                    | 2.5                     | Pass          |
|             |             | C<br>(Middle of String) | 1       | 139.1    | 2.7      | 89.3     | 1.444                    | 2.5                     | Pass          |
|             |             |                         | 2       | 183.5    | 2.6      | 47.7     | 1.037                    | 2.5                     | Pass          |
|             |             |                         | 3       | 173.5    | 2.6      | 3.1      | 0.510                    | 2.5                     | Pass          |
|             |             |                         | 4       | 27.3     | 2.7      | 54.2     | 2.167                    | 2.5                     | Pass          |
|             |             |                         | 5       | 100.8    | 2.6      | 18.3     | 0.319                    | 2.5                     | Pass          |
|             | 2           | A<br>(Start of String)  | 1       | 19.2     | 7.0      | 132.1    | 1.413                    | 2.5                     | Pass          |
|             |             |                         | 2       | 23.1     | 7.1      | 136.7    | 1.640                    | 2.5                     | Pass          |
|             |             |                         | 3       | 23.1     | 7.3      | 135.1    | 1.641                    | 2.5                     | Pass          |
|             |             |                         | 4       | 22.5     | 7.3      | 131.4    | 1.671                    | 2.5                     | Pass          |
|             |             |                         | 5       | 19.5     | 7.4      | 115.3    | 2.074                    | 2.5                     | Pass          |

|  |   |                         |   |      |      |       |       |     |      |
|--|---|-------------------------|---|------|------|-------|-------|-----|------|
|  |   | B<br>(End of String)    | 1 | 13.8 | 7.3  | 103.5 | 2.063 | 2.5 | Pass |
|  |   |                         | 2 | 24.9 | 7.6  | 176.4 | 2.167 | 2.5 | Pass |
|  |   |                         | 3 | 26.3 | 7.3  | 167.9 | 1.950 | 2.5 | Pass |
|  |   |                         | 4 | 21.2 | 7.3  | 133.5 | 1.765 | 2.5 | Pass |
|  |   |                         | 5 | 23.1 | 7.4  | 149.4 | 2.208 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 24.4 | 7.3  | 151.3 | 2.356 | 2.5 | Pass |
|  |   |                         | 2 | 23.8 | 7.5  | 163.2 | 2.063 | 2.5 | Pass |
|  |   |                         | 3 | 23.3 | 7.5  | 159.6 | 2.177 | 2.5 | Pass |
|  |   |                         | 4 | 23.5 | 7.5  | 159.0 | 2.043 | 2.5 | Pass |
|  |   |                         | 5 | 22.8 | 7.6  | 159.1 | 2.156 | 2.5 | Pass |
|  | 3 | A<br>(Start of String)  | 1 | 21.7 | 14.7 | 263.1 | 1.971 | 2.5 | Pass |
|  |   |                         | 2 | 8.5  | 14.9 | 119.0 | 1.558 | 2.5 | Pass |
|  |   |                         | 3 | 23.5 | 14.1 | 252.0 | 1.826 | 2.5 | Pass |
|  |   |                         | 4 | 21.2 | 14.5 | 252.1 | 1.805 | 2.5 | Pass |
|  |   |                         | 5 | 23.8 | 14.1 | 253.0 | 1.073 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 24.9 | 14.0 | 264.3 | 1.031 | 2.5 | Pass |
|  |   |                         | 2 | 19.0 | 14.5 | 133.1 | 1.393 | 2.5 | Pass |
|  |   |                         | 3 | 19.6 | 14.4 | 137.6 | 1.321 | 2.5 | Pass |
|  |   |                         | 4 | 23.8 | 14.1 | 254.5 | 2.270 | 2.5 | Pass |
|  |   |                         | 5 | 22.6 | 14.5 | 268.1 | 2.002 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 19.7 | 14.1 | 219.1 | 1.630 | 2.5 | Pass |
|  |   |                         | 2 | 21.8 | 14.7 | 282.5 | 1.661 | 2.5 | Pass |
|  |   |                         | 3 | 6.4  | 14.9 | 55.4  | 1.341 | 2.5 | Pass |
|  |   |                         | 4 | 6.5  | 14.9 | 57.2  | 1.310 | 2.5 | Pass |
|  |   |                         | 5 | 23.2 | 14.5 | 274.7 | 1.950 | 2.5 | Pass |
|  | 4 | A<br>(Start of String)  | 1 | 20.6 | 7.9  | 143.4 | 1.744 | 2.5 | Pass |
|  |   |                         | 2 | 20.4 | 7.4  | 112.7 | 1.269 | 2.5 | Pass |
|  |   |                         | 3 | 19.8 | 7.4  | 117.5 | 1.568 | 2.5 | Pass |
|  |   |                         | 4 | 23.1 | 7.3  | 130.6 | 1.042 | 2.5 | Pass |
|  |   |                         | 5 | 20.7 | 7.9  | 136.2 | 1.723 | 2.5 | Pass |
|  |   | B<br>(End of String)    | 1 | 23.2 | 7.5  | 126.3 | 1.228 | 2.5 | Pass |
|  |   |                         | 2 | 21.2 | 7.7  | 137.7 | 1.434 | 2.5 | Pass |
|  |   |                         | 3 | 20.6 | 7.6  | 124.6 | 1.847 | 2.5 | Pass |
|  |   |                         | 4 | 10.4 | 7.6  | 77.7  | 1.723 | 2.5 | Pass |
|  |   |                         | 5 | 24.9 | 7.4  | 140.3 | 1.331 | 2.5 | Pass |
|  |   | C<br>(Middle of String) | 1 | 24.1 | 7.4  | 148.9 | 1.517 | 2.5 | Pass |
|  |   |                         | 2 | 13.3 | 7.2  | 86.3  | 1.589 | 2.5 | Pass |
|  |   |                         | 3 | 18.6 | 7.3  | 110.3 | 1.527 | 2.5 | Pass |
|  |   |                         | 4 | 23.0 | 7.6  | 149.8 | 1.217 | 2.5 | Pass |
|  |   |                         | 5 | 23.3 | 7.3  | 138.2 | 1.093 | 2.5 | Pass |

Note:

- After evaluation, when C4=150nF, the test conditions are the worst.

#### 4.4 UNWANTED TRIPPING TESTS

The verification of Resistance to Environmental Noise Test has been measured according to the chapter 30 of the standard.

A representative device shall be tested in accordance with Figures 30.1 – 30.5, using the values in Table 30.1 and the legend in Figure 29.1. The device shall not trip after being tested under each of the loading and sourcing conditions as described in this Section. When tripping occurs, an additional five representative devices of the rating under test shall be tested and shall not trip.

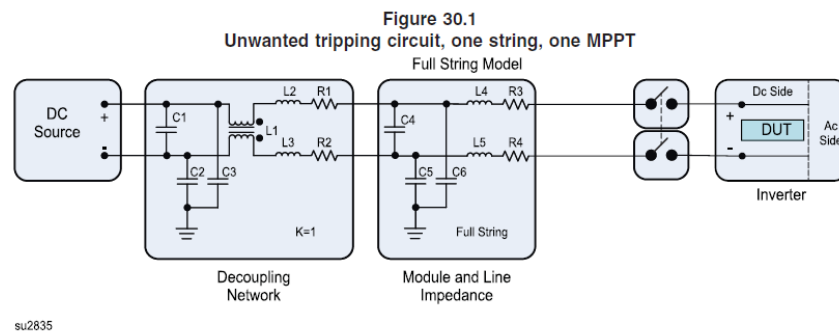
##### 4.4.1 LOADING CONDITION I – INVERTERS, CONVERTERS, AND CHARGE CONTROLLERS

The DUT shall be installed 0.6 m (2.0 ft) from the load device. With a PV power source operating at rated power and when agreed upon by all concerned parties, with one MPP-voltage from Table 29.2 representative of the unit under test, the load device shall be turned on for five minutes minimum and off for five minutes minimum. This sequence shall be repeated for a total of three operations.

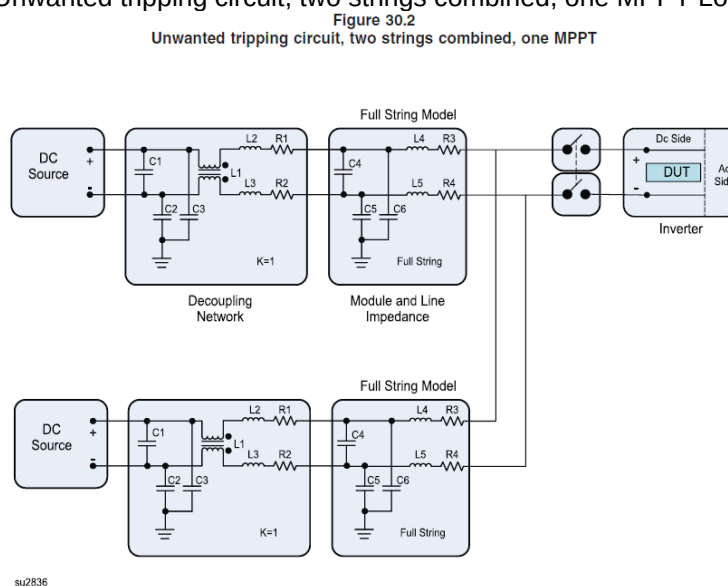
The DUT shall NOT indicate an arc fault.

Model Type:

- 1) Figure 30.1: Unwanted tripping circuit, one string, one MPPT Location: Close to DUT



- 2) Figure 30.2: Unwanted tripping circuit, two strings combined, one MPPT Location: Close to DUT



**Test Result: PASS**

| Test Condition       | Cap.(C4) (nF) | Attempts | Result:                   |
|----------------------|---------------|----------|---------------------------|
| one string, one MPPT | C4=150nF      | 1        | Not indicate an arc fault |
|                      |               | 2        | Not indicate an arc fault |
|                      |               | 3        | Not indicate an arc fault |
|                      |               | 4        | Not indicate an arc fault |
|                      |               | 5        | Not indicate an arc fault |

#### 4.4.2 LOADING CONDITION II – DC SWITCH OPERATION

The DUT shall be installed in a test circuit with a PV power source and an inverter, converter, or charge controller. A disconnect switch external to the inverter shall be installed two feet from the device under test. With the PV power source set up to supply the rated power of the device under test, the disconnect switch shall be operated “on” and then after the inverter has been exporting power for 30 seconds, moved to the “off” position. This sequence shall be repeated three times, with five minutes “off” time between each sequence.

The DUT shall NOT indicate an arc fault.

**Test Result: PASS**

| Test Condition       | Cap.(C4) (nF) | Attempts | Result:                   |
|----------------------|---------------|----------|---------------------------|
| one string, one MPPT | C4=150nF      | 1        | Not indicate an arc fault |
|                      |               | 2        | Not indicate an arc fault |
|                      |               | 3        | Not indicate an arc fault |
|                      |               | 4        | Not indicate an arc fault |
|                      |               | 5        | Not indicate an arc fault |

#### 4.4.3 LOADING CONDITION III – IRRADIANCE STEP CHANGES

The DUT shall be installed in a test circuit with a PV power source and an inverter as shown in Figures 30.6A and 30.6B. A mechanical disconnect switch shall be installed on one half of the PV strings connected to the inverter. With the PV power source operating in a manner to supply the rated power of the device under test, the disconnect switch attached to half the PV source circuits shall be operated “on” and then after 30 seconds, “off”. This sequence shall be repeated three times, with five minutes “off” time between each sequence. If a single power source is used this test must be performed by rapidly decreasing the supplied current to at least ½ of the test current in less than 5 seconds. After 30 seconds return to full initial current in less than 5 seconds. If using a module or array this may be achieved with a cover over the array or module. See Figure 30.1.

The DUT shall NOT indicate an arc fault.

**Test Result: PASS**

| Test Condition       | Cap.(C4) (nF) | Attempts | Result:                   |
|----------------------|---------------|----------|---------------------------|
| one string, one MPPT | C4=150nF      | 1        | Not indicate an arc fault |
|                      |               | 2        | Not indicate an arc fault |
|                      |               | 3        | Not indicate an arc fault |
|                      |               | 4        | Not indicate an arc fault |
|                      |               | 5        | Not indicate an arc fault |

#### 4.5 NORMAL TEMPERATURE TEST

The verification of Resistance to Environmental Noise Test has been measured according to the chapter 31 of the standard.

When carrying rated current and with rated voltage applied, a device shall not attain a temperatura at any point that is sufficiently high to:

- a) Constitute an increased risk of fire;
- b) Affect injuriously any materials used in the device; or
- c) Exhibit greater rises in temperature at specific points than indicated in Table 31.1, based on an assumed average ambient temperature in normal service of 25°C (77°F).

Coil or winding temperatures are to be measured by thermocouples unless access cannot be gained for mounting a thermocouple (for example, a coil enclosed in sealing compound) or unless the coil wrap includes thermal insulation or more than two layers (0.8 mm or 1/32 inch maximum) of cotton, paper, rayon, or the like. At a point on the surface of a coil where the temperature is affected by an external source of heat, the temperature rise measured by means of a thermocouple may be 10°C (18°F) more than the indicated maximum, provided that the temperature rise of the coil, as measured by the resistance method is no more than that specified in Table 31.1.

Ambient air is to be at any convenient temperature within the range of 20 – 30°C (68 – 86°F).

**Table 31.1**  
**Maximum acceptable temperature rises**

| Material and components                                                                                                                                                                                                                               | °C  | (°F)  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|
| Wire insulation or insulating tubing                                                                                                                                                                                                                  | 35  | (63)  |
| Electrical tape                                                                                                                                                                                                                                       | 55  | (99)  |
| Varnish-cloth insulation                                                                                                                                                                                                                              | 60  | (108) |
| Fiber employed as electrical insulation                                                                                                                                                                                                               | 65  | (117) |
| Phenolic composition or melamine <sup>a</sup>                                                                                                                                                                                                         | 125 | (198) |
| Urea composition <sup>a</sup>                                                                                                                                                                                                                         | 75  | (108) |
| Other insulating materials <sup>a</sup>                                                                                                                                                                                                               | –   | –     |
| <sup>a</sup> The acceptability of insulating materials shall be determined with respect to properties – such as flammability, arc resistance, relative or generic temperature indices, and the like – based on the temperature rise plus 25°C (45°F). |     |       |

**Test Result: PASS**

| THERMOCOUPLE LOCATION |          | TC. # | MEASURED TEMPERATURE | MAXIMUM ALLOWED TEMPERATURE (°C) |
|-----------------------|----------|-------|----------------------|----------------------------------|
| 1                     | IGBT 1   | CH 1  | 83.5                 | 175                              |
| 2                     | IGBT 14  | CH 2  | 67.6                 | 175                              |
| 3                     | IGBT 11  | CH 3  | 84.2                 | 175                              |
| 4                     | IGBT 15  | CH 4  | 70.2                 | 175                              |
| 5                     | IGBT Q16 | CH 5  | 54.3                 | 175                              |
| 6                     | IGBT Q9  | CH 6  | 54.7                 | 175                              |
| 7                     | IGBT 20  | CH 7  | 66.9                 | 175                              |

| THERMOCOUPLE LOCATION |                                             | TC. # | MEASURED TEMPERATURE | MAXIMUM ALLOWED TEMPERATURE (°C) |
|-----------------------|---------------------------------------------|-------|----------------------|----------------------------------|
| 8                     | IGBT 5                                      | CH 8  | 62.9                 | 175                              |
| 9                     | IGBT Q2                                     | CH 9  | 54.0                 | 175                              |
| 10                    | IGBT Q3                                     | CH 10 | 54.2                 | 175                              |
| 11                    | PCB board near the positive terminal of BAT | CH 11 | 55.1                 | 130                              |
| 12                    | BAT positive magnetic ring SN2              | CH 12 | 55.7                 | --                               |
| 13                    | Capacitor(C410)                             | CH 13 | 56.7                 | 105                              |
| 14                    | Capacitor(C393)                             | CH 14 | 54.7                 | 105                              |
| 15                    | Capacitor(C421)                             | CH 15 | 57.1                 | 105                              |
| 16                    | Control chip U9                             | CH 16 | 60.0                 | 125                              |
| 17                    | Temperature probe BAT side                  | CH 17 | 65.5                 | 200                              |
| 18                    | Temperature probe INV side                  | CH 18 | 71.8                 | 200                              |
| 19                    | Inv inductance                              | CH 19 | 58.5                 | 155                              |
| 20                    | PV boost inductance                         | CH 20 | 93.1                 | 155                              |
| 21                    | INV optocoupler (PC38)                      | CH 21 | 69.7                 | 150                              |
| 22                    | PCB board near INVP                         | CH 22 | 68.2                 | 130                              |
| 23                    | Relay K1                                    | CH 23 | 56.9                 | 85                               |
| 24                    | Inv Capacitor(C373)                         | CH 24 | 70.3                 | 105                              |
| 25                    | Magnetic ring (SN1)                         | CH 25 | 69.0                 | --                               |
| 26                    | Bus Capacitor                               | CH 26 | 65.2                 | 105                              |
| 27                    | INV1 positive cable                         | CH 27 | 67.0                 | 105                              |
| 28                    | BOOST capacitor C375                        | CH 28 | 61.3                 | 105                              |
| 29                    | inductance(L2)                              | CH 29 | 60.4                 | 155                              |
| 30                    | BOOST optocoupler PC16                      | CH 30 | 61.9                 | 130                              |
| 31                    | BAT positive cable                          | CH 31 | 65.3                 | 105                              |
| 32                    | BOOST negative cable                        | CH 32 | 61.1                 | 105                              |
| 33                    | Battery side optocoupler PC32               | CH 33 | 54.6                 | 130                              |
| 34                    | discharge resistance                        | CH 34 | 61.2                 | --                               |
| 35                    | Capacitor(U3)                               | CH 35 | 60.2                 | 105                              |
| 36                    | Capacitor(U11)                              | CH 36 | 57.8                 | 105                              |
| 37                    | Fan(internal)                               | CH 37 | 58.3                 | 70                               |
| 38                    | Relay RLY1                                  | CH 38 | 60.4                 | 85                               |
| 39                    | Transformer TR3                             | CH 39 | 63.1                 | 130                              |
| 40                    | Battery side optocoupler PC31               | CH 40 | 55.1                 | 100                              |
| 41                    | Capacitor(C22)                              | CH 41 | 63.4                 | 105                              |
| 42                    | Transformer TR4                             | CH 42 | 74.6                 | 130                              |
| 43                    | Transformer TR1                             | CH 43 | 61.4                 | 130                              |
| 44                    | inductance(L3)                              | CH 44 | 61.8                 | 155                              |
| 45                    | Transistor SH2                              | CH 45 | 58.4                 | 125                              |
| 46                    | PCB board near the chip                     | CH 46 | 65.0                 | 130                              |
| 47                    | Transistor U9                               | CH 47 | 61.2                 | 125                              |
| 48                    | Transistor SH1                              | CH 48 | 70.0                 | 125                              |
| 49                    | Capacitor(C78)                              | CH 49 | 63.4                 | 105                              |
| 50                    | Relay RLY1                                  | CH 50 | 75.1                 | 85                               |
| 51                    | Transformer coil L16                        | CH 51 | 65.0                 | 130                              |
| 52                    | Leakage current magnetic ring SN7           | CH 52 | 63.2                 | --                               |
| 53                    | Transformer coil L16                        | CH 53 | 68.0                 | 130                              |

| THERMOCOUPLE LOCATION |                                 | TC. # | MEASURED TEMPERATURE | MAXIMUM ALLOWED TEMPERATURE (°C) |
|-----------------------|---------------------------------|-------|----------------------|----------------------------------|
| 54                    | Relay RLY2                      | CH 54 | 74.3                 | 85                               |
| 55                    | Capacitor(C75)                  | CH 55 | 60.6                 | 105                              |
| 56                    | Relay RLY3                      | CH 56 | 60.3                 | 85                               |
| 57                    | Magnetic ring (SN11)            | CH 57 | 57.7                 | --                               |
| 58                    | Magnetic ring (SN12)            | CH 58 | 59.3                 | --                               |
| 59                    | Magnetic ring (SN10)            | CH 59 | 58.6                 | --                               |
| 60                    | Magnetic ring (SN13)            | CH 60 | 59.5                 | --                               |
| 61                    | Capacitor(C66)                  | CH 61 | 57.1                 | 105                              |
| 62                    | inductance(L2)                  | CH 62 | 57.1                 | 105                              |
| 63                    | Capacitor(C68)                  | CH 63 | 58.2                 | 105                              |
| 64                    | inductance(L1)                  | CH 64 | 58.1                 | 105                              |
| 65                    | Varistor (RV1)                  | CH 1  | 56.7                 | 85                               |
| 66                    | Capacitor(C15)                  | CH 2  | 58.3                 | 105                              |
| 67                    | PV switch (internal)            | CH 3  | 56.2                 | 85                               |
| 68                    | Capacitor(C7)                   | CH 4  | 56.7                 | 105                              |
| 69                    | Capacitor(C64)                  | CH 5  | 57.6                 | 105                              |
| 70                    | Capacitor(C62)                  | CH 6  | 57.2                 | 105                              |
| 71                    | Varistor (RV8)                  | CH 7  | 56.7                 | 85                               |
| 72                    | Discharge diode (GDT1)          | CH 8  | 57.0                 | 85                               |
| 73                    | Cable GEN-L1                    | CH 9  | 57.0                 | 105                              |
| 74                    | Transformer coil L15            | CH 10 | 75.4                 | 130                              |
| 75                    | Current detection TR2           | CH 11 | 58.5                 | --                               |
| 76                    | Capacitor(C34)                  | CH 12 | 62.0                 | 105                              |
| 77                    | Magnetic ring (SN9)             | CH 13 | 60.8                 | --                               |
| 78                    | inductance(L3)                  | CH 14 | 59.7                 | 105                              |
| 79                    | Transformer coil L15            | CH 15 | 63.0                 | 130                              |
| 80                    | Relay RLY5                      | CH 16 | 62.3                 | 85                               |
| 81                    | Right side inside the shell     | CH 17 | 52.8                 | 90                               |
| 82                    | Relay K3                        | CH 18 | 67.7                 | 85                               |
| 83                    | Fuse F10                        | CH 19 | 59.9                 | 85                               |
| 84                    | Left side inside the shell      | CH 20 | 55.7                 | 90                               |
| 85                    | Fuse F8                         | CH 21 | 58.9                 | 85                               |
| 86                    | Current detection TR1           | CH 22 | 58.4                 | --                               |
| 87                    | Upper side inside the shell     | CH 23 | 53.5                 | 90                               |
| 88                    | Relay K2                        | CH 24 | 72.2                 | 85                               |
| 89                    | Grid switch 2                   | CH 25 | 57.7                 | 70                               |
| 90                    | Load switch 2                   | CH 26 | 51.6                 | 70                               |
| 91                    | Grid cable(L1)                  | CH 27 | 58.5                 | 105                              |
| 92                    | Grid switch 1                   | CH 28 | 57.9                 | 70                               |
| 93                    | Load switch 1                   | CH 29 | 53.4                 | 70                               |
| 94                    | Grid cable(L2)                  | CH 30 | 57.5                 | 105                              |
| 95                    | Load cable(L1)                  | CH 31 | 55.5                 | 105                              |
| 96                    | Load cable(L2)                  | CH 32 | 53.4                 | 105                              |
| 97                    | Inner bottom of shell           | CH 33 | 47.7                 | 90                               |
| 98                    | PV terminal                     | CH 34 | 53.1                 | 110                              |
| 99                    | communication port              | CH 35 | 53.5                 | 90                               |
| 100                   | Battery switch                  | CH 36 | 54.0                 | 70                               |
| 101                   | Internal ambient temperature of | CH 37 | 61.1                 | 90                               |

| THERMOCOUPLE LOCATION                                                                                         |                             | TC. # | MEASURED TEMPERATURE | MAXIMUM ALLOWED TEMPERATURE (°C) |
|---------------------------------------------------------------------------------------------------------------|-----------------------------|-------|----------------------|----------------------------------|
|                                                                                                               | the machine                 |       |                      |                                  |
| 102                                                                                                           | PV cable                    | CH 38 | 55.3                 | 105                              |
| 103                                                                                                           | Bat cable                   | CH 39 | 53.4                 | 105                              |
| 104                                                                                                           | Top inside the shell        | CH 40 | 51.2                 | 90                               |
| 105                                                                                                           | inductance enclosure(left)  | CH 41 | 53.1                 | 105                              |
| 106                                                                                                           | External heat sink          | CH 42 | 58.4                 | --                               |
| 107                                                                                                           | inductance enclosure(right) | CH 43 | 49.4                 | 105                              |
| 108                                                                                                           | enclosure temperature       | CH 44 | 52.3                 | 90                               |
| 109                                                                                                           | ambient temperature         | CH 45 | 45.5                 | --                               |
| 110                                                                                                           | External fan (top)          | CH 46 | 49.9                 | 70                               |
| 111                                                                                                           | External fan (lower)        | CH 47 | 49.9                 | 70                               |
| 112                                                                                                           | Screen                      | CH 48 | 51.1                 | 90                               |
| 113                                                                                                           | handle                      | CH 51 | 49.0                 | 90                               |
| 114                                                                                                           | start button                | CH 53 | 48.0                 | 90                               |
| 115                                                                                                           | PV switch(external)         | CH 54 | 46.1                 | 85                               |
| <b>Note: Considering the maximum operating temperature of the inverter, this test is carried out at 45 °C</b> |                             |       |                      |                                  |

#### 4.6 DIELECTRIC VOLTAGE-WITHSTAND TEST

The verification of Resistance to Environmental Noise Test has been measured according to the chapter 35 of the standard.

The insulation and spacings shall withstand the application of a test voltage equal to two times the maximum rated voltage plus 1000 V between:

- a) Line and load with the device open for a device that includes a circuit interrupting contact;
- b) Line and load with the device tripped for a device that includes a circuit interrupting contact;
- c) Live parts and parts that are grounded, including a ground terminal;
- d) Live parts and accessible metal parts including an enclosure of insulating material wrapped in metal foil. See 35.7.

For a device deriving power from a commercial light or power source, the test voltage shall also be applied between the ac circuit and dc circuits, and between the ac circuit and parts that are grounded.

For a device deriving power from a commercial light or power source, the test voltage shall also be applied between the ac circuit and dc circuits, and between the ac circuit and parts that are grounded.

In order to determine compliance with the provisions of 35.1, the insulation and spacings are to be subjected to application of a test potential increased from zero to the values specified and maintained for a period of one minute. The increase in the applied potential is to be at a substantially uniform rate and as rapid as is consistent with the value of the applied potential being correctly indicated by the voltmeter. The source of this test voltage shall be in accordance with (a) or (b) as follows:

- a) For ac circuits, a 60 hertz essentially sinusoidal potential;
- b) For a dc circuit, or between ac and dc circuits, a direct-current potential. It shall be permissible to use an alternating-current potential for these circuits. The ac RMS potential value is to be the required dc dielectric potential divided by 1.414.

**Test Result: PASS**

| <i><b>TEST LOCATION</b></i>             | <i><b>APPLIED VOLTAGE</b></i> | <i><b>PASS / FAIL</b></i> |
|-----------------------------------------|-------------------------------|---------------------------|
| AC Output circuit to dead metal         | 1480Vac(2200Vdc)              | PASS                      |
| AC Output circuit to communication port | 1480Vac(2200Vdc)              | PASS                      |
| AC Output circuit to Battery terminal   | 1480Vac(2200Vdc)              | PASS                      |
| DC Input circuit to dead metal          | 2000Vdc                       | PASS                      |
| DC Input circuit to Battery terminal    | 2000Vdc                       | PASS                      |
| Battery terminal to dead metal          | 1120Vdc                       | PASS                      |

#### 4.7 RESISTANCE TO ENVIRONMENTAL NOISE TEST

The verification of Resistance to Environmental Noise Test has been measured according to the chapter 42 of the standard.

The same representative device shall be tested for all of the applicable tests as tabulated in Table 24.1 and described in 42.2 – 42.7.

Except as indicated in 42.1.4, the device shall not trip as a result of the electromagnetic event.

It shall be permissible for a device to trip as a result of any electromagnetic event, providing all of the following conditions are met:

- a) The device automatically restores power after conducting an acceptable self-test;
- b) The self-test system that allows restoration of power after an electromagnetic event also prevents the device from resetting automatically after tripping as a result of an arc fault; and
- c) If the self-test system employs a programmable circuit component such as a microprocessor, it shall be included in the investigation required by Section 13, Programmable Circuit Components.

It shall be permissible for a device to trip during the Voltage dips, short interruptions and voltage variations immunity test if the device complies with 42.1.4 or 42.7.2.

##### Test Record:

| Item | Content                                                                                                                                                                               | Verdict<br>(Yes or No) |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| a    | The device automatically restores power after conducting an acceptable self-test;                                                                                                     | Yes                    |
| b    | The self-test system that allows restoration of power after an electromagnetic event also prevents the device from resetting automatically after tripping as a result of an arc fault | Yes                    |
| c    | If the self-test system employs a programmable circuit component such as a microprocessor, it shall be included in the investigation required by Section 13.                          | Yes                    |

After all of the tests involving the electromagnetic events, the representative device shall be tested in accordance with 29.1 using any two of the test conditions specified in Table 29.2, and shall trip as required.

**Test Result: PASS**

One string, one mppt:

| Tem<br>.(°C) | Test<br>No. | Location                   | Attempt | Varc<br>(V) | Iarc (A) | Parc<br>(W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pass/<br>Fail |
|--------------|-------------|----------------------------|---------|-------------|----------|-------------|--------------------------|-------------------------|---------------|
| 25°C         | 2           | A<br>(Start of<br>String)  | 1       | 19.6        | 7.2      | 125.3       | 1.775                    | 2.5                     | Pass          |
|              |             |                            | 2       | 113.6       | 7.3      | 94.4        | 0.970                    | 2.5                     | Pass          |
|              |             |                            | 3       | 21.0        | 7.1      | 127.8       | 1.991                    | 2.5                     | Pass          |
|              |             |                            | 4       | 22.9        | 7.3      | 133.8       | 1.568                    | 2.5                     | Pass          |
|              |             |                            | 5       | 21.3        | 7.4      | 124.0       | 1.857                    | 2.5                     | Pass          |
|              |             | B<br>(End of<br>String)    | 1       | 20.4        | 7.1      | 116.7       | 1.671                    | 2.5                     | Pass          |
|              |             |                            | 2       | 20.8        | 7.0      | 117.7       | 1.764                    | 2.5                     | Pass          |
|              |             |                            | 3       | 22.1        | 7.0      | 116.5       | 1.898                    | 2.5                     | Pass          |
|              |             |                            | 4       | 19.8        | 7.0      | 123.6       | 2.012                    | 2.5                     | Pass          |
|              |             |                            | 5       | 21.3        | 7.2      | 136.3       | 1.537                    | 2.5                     | Pass          |
|              |             | C<br>(Middle of<br>String) | 1       | 13.9        | 7.4      | 55.6        | 1.661                    | 2.5                     | Pass          |
|              |             |                            | 2       | 20.9        | 7.2      | 133.6       | 2.053                    | 2.5                     | Pass          |
|              |             |                            | 3       | 19.6        | 7.2      | 119.3       | 1.867                    | 2.5                     | Pass          |
|              |             |                            | 4       | 21.6        | 7.0      | 119.6       | 1.341                    | 2.5                     | Pass          |
|              |             |                            | 5       | 21.7        | 7.1      | 134.1       | 1.775                    | 2.5                     | Pass          |

Note:

1. After evaluation, when C4=150nF, the test conditions are the worst.

One string, one mppt :

| Tem<br>.(°C) | Test<br>No. | Location                   | Attempt | Varc<br>(V) | Iarc (A) | Parc<br>(W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pass/<br>Fail |
|--------------|-------------|----------------------------|---------|-------------|----------|-------------|--------------------------|-------------------------|---------------|
| 25°C         | 3           | A<br>(Start of<br>String)  | 1       | 22.8        | 14.0     | 264.4       | 1.444                    | 2.5                     | Pass          |
|              |             |                            | 2       | 21.7        | 14.2     | 268.7       | 1.764                    | 2.5                     | Pass          |
|              |             |                            | 3       | 22.9        | 14.0     | 265.4       | 2.012                    | 2.5                     | Pass          |
|              |             |                            | 4       | 21.7        | 14.1     | 247.9       | 1.764                    | 2.5                     | Pass          |
|              |             |                            | 5       | 22.3        | 14.0     | 254.6       | 1.733                    | 2.5                     | Pass          |
|              |             | B<br>(End of<br>String)    | 1       | 22.7        | 14.5     | 261.4       | 1.052                    | 2.5                     | Pass          |
|              |             |                            | 2       | 30.2        | 14.0     | 238.5       | 1.661                    | 2.5                     | Pass          |
|              |             |                            | 3       | 124.5       | 14.1     | 197.9       | 1.177                    | 2.5                     | Pass          |
|              |             |                            | 4       | 22.7        | 14.6     | 256.1       | 1.382                    | 2.5                     | Pass          |
|              |             |                            | 5       | 20.3        | 14.7     | 258.7       | 1.775                    | 2.5                     | Pass          |
|              |             | C<br>(Middle of<br>String) | 1       | 23.3        | 14.5     | 284.7       | 1.310                    | 2.5                     | Pass          |
|              |             |                            | 2       | 23.8        | 14.8     | 293.7       | 2.053                    | 2.5                     | Pass          |
|              |             |                            | 3       | 23.3        | 14.8     | 287.1       | 2.136                    | 2.5                     | Pass          |
|              |             |                            | 4       | 107.8       | 14.4     | 185.1       | 1.506                    | 2.5                     | Pass          |
|              |             |                            | 5       | 25.3        | 14.1     | 281.6       | 1.351                    | 2.5                     | Pass          |

Note:

1. After evaluation, when C4=150nF, the test conditions are the worst.

|                                                                                   |                                   |               |
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#### **4.8 VOLTAGE SURGE TEST**

Devices deriving their power from a photovoltaic DC source and intended for use in stand-alone (not grid-connected) power systems shall be subjected to the tests of 27.2 and 27.3 with the surges applied to the PV DC input of the device.

Devices deriving their power from a commercial light or power source and utility-interactive equipment shall be subjected to the surge tests described in the Standard for Interconnecting Distributed Resources With Electric Power Systems, IEEE 1547. The surges shall be applied to the AC input of the device.

In addition to the test of 55.2, devices deriving their power from a commercial light or power source and utility-interactive equipment shall also be subjected to the tests of 27.2 and 27.3 with the surges applied to the PV DC input of the device.

**Test Result: PASS**

For Inverter: See report: 01GZES240100172601 Amendment 01 of IEEE 1547

|                                                                                   |                                   |               |
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#### **4.9 OVERVOLTAGE, OVERLOAD, ENDURANCE, ABNORMAL OPERATIONS, AND SHORT CIRCUIT TESTS**

The verification of Resistance to Environmental Noise Test has been measured according to the chapter 56 of the standard.

Switching devices in the power conversion circuits of inverters, converters, and charge controllers that are subjected to the Abnormal Tests, as applicable, of the Standard for Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources, UL 1741, or the Standard for Power Converters for Use in Photovoltaic Power Systems – Part 1: General Requirements, UL 62109-1, need not be subjected to the Overvoltage, Overload, Endurance, Abnormal Operations, and Short Circuit Tests from Table 54.1, provided that after the Abnormal Tests of UL 1741 or UL 62109-1, each representative device complies with 14.2.1 and the applicable tests of Section 29, Arc Fault Detection Tests. See 56.2.

When applying the requirements of 56.1, the applicable tests of Section 29, Arc Fault Detection Tests, shall not be conducted after an Abnormal Test if all of the following conditions are met:

- a) The sample is no longer functional, that is, has no inverter output; and
- b) The sample provides an indication to the user that it has failed.

When conducting the applicable tests of Section 29, Arc Fault Detection Tests, the device shall interrupt the arcing as described in Table 29.2 for the conditions of Test No. 2 and 3 at room ambient.

Redundant interrupting or shunting contact devices that supplement power conversion circuits of inverters, converters, and charge controllers that do not operate as part of the power conversion electronics shall be subjected to the Overvoltage, Overload, Endurance, and Short Circuit tests from Table 54.1.

**Test Result:** The inverter has completed the certification test of UL1741:2021.Check after abnormal test:  
PASS, one string, one MPPT

| Tem .(°C) | Tes t No. | Location                                                               | Attempt | Varc (V) | Iarc (A) | Parc (W) | Trip Time measured(s) | Trip Time limited(s) | Pas s/ Fail |  |  |
|-----------|-----------|------------------------------------------------------------------------|---------|----------|----------|----------|-----------------------|----------------------|-------------|--|--|
| 25°C      | 2         | A<br>(Start of String)                                                 | 1       | 21.8     | 7.2      | 137.2    | 1.857                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 2       | 18.2     | 7.0      | 99.7     | 2.311                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 3       | 21.4     | 7.4      | 140.1    | 1.971                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 4       | 22.9     | 7.1      | 137.3    | 1.455                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 5       | 20.7     | 7.2      | 132.0    | 2.063                 | 2.5                  | Pass        |  |  |
|           |           | B<br>(End of String)                                                   | 1       | 20.3     | 7.0      | 117.8    | 1.733                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 2       | 21.1     | 7.3      | 129.2    | 1.764                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 3       | 22.2     | 7.2      | 136.1    | 1.723                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 4       | 21.0     | 7.0      | 124.3    | 1.847                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 5       | 21.1     | 7.2      | 122.6    | 1.527                 | 2.5                  | Pass        |  |  |
|           |           | C<br>(Middle of String)                                                | 1       | 22.0     | 7.4      | 150.6    | 2.022                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 2       | 22.5     | 7.4      | 154.3    | 1.940                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 3       | 16.8     | 7.1      | 86.4     | 2.063                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 4       | 20.8     | 7.1      | 126.3    | 1.568                 | 2.5                  | Pass        |  |  |
|           |           |                                                                        | 5       | 27.7     | 7.0      | 121.6    | 1.682                 | 2.5                  | Pass        |  |  |
|           |           | Note:                                                                  |         |          |          |          |                       |                      |             |  |  |
|           |           | 1. After evaluation, when C4=150nF, the test conditions are the worst. |         |          |          |          |                       |                      |             |  |  |

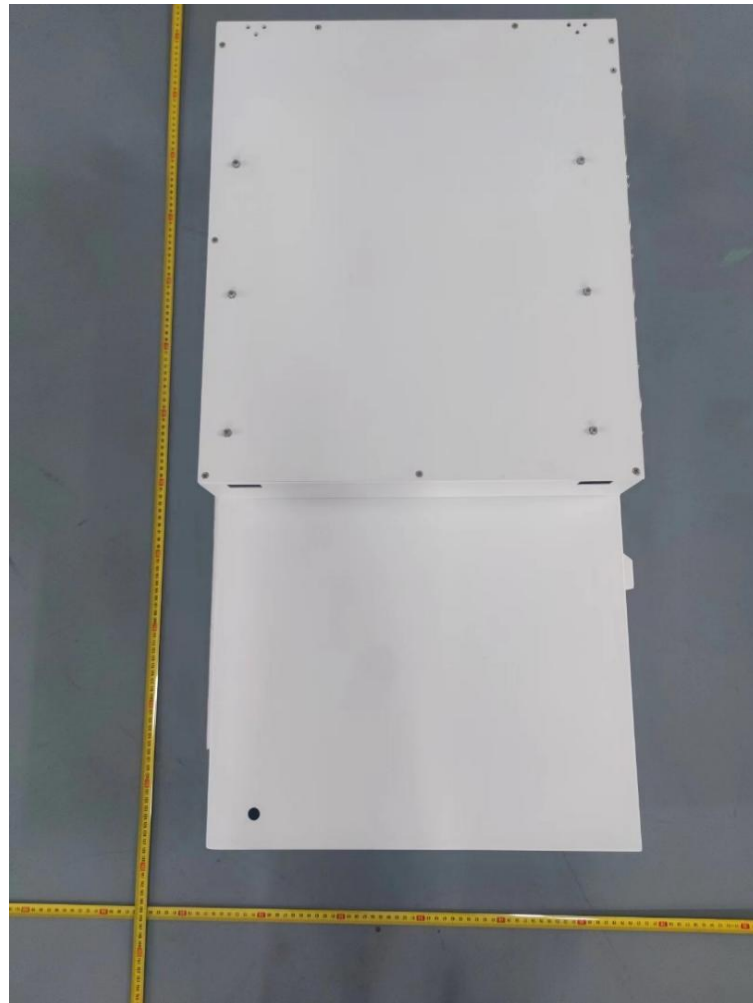
| Tem<br>.(°C) | Tes<br>t<br>No. | Location                                                               | Attempt | Varc<br>(V) | Iarc (A) | Parc<br>(W) | Trip Time<br>measured(s) | Trip Time<br>limited(s) | Pas<br>s/<br>Fail |  |  |
|--------------|-----------------|------------------------------------------------------------------------|---------|-------------|----------|-------------|--------------------------|-------------------------|-------------------|--|--|
| 25°C         | 3               | A<br>(Start of<br>String)                                              | 1       | 21.7        | 14.7     | 242.2       | 1.661                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 2       | 22.7        | 14.1     | 268.3       | 1.434                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 3       | 23.3        | 14.0     | 222.1       | 2.177                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 4       | 22.0        | 14.2     | 274.2       | 2.063                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 5       | 21.4        | 14.3     | 251.0       | 1.444                    | 2.5                     | Pass              |  |  |
|              |                 | B<br>(End of<br>String)                                                | 1       | 24.2        | 14.0     | 232.6       | 1.031                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 2       | 22.3        | 14.1     | 213.6       | 1.960                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 3       | 7.4         | 15.1     | 94.1        | 1.878                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 4       | 24.0        | 14.0     | 260.4       | 1.651                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 5       | 22.5        | 14.6     | 275.2       | 1.362                    | 2.5                     | Pass              |  |  |
|              |                 | C<br>(Middle of<br>String)                                             | 1       | 21.2        | 14.6     | 264.4       | 2.053                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 2       | 21.7        | 14.6     | 270.6       | 1.971                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 3       | 19.9        | 15.0     | 245.1       | 1.795                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 4       | 18.9        | 15.1     | 233.2       | 1.991                    | 2.5                     | Pass              |  |  |
|              |                 |                                                                        | 5       | 122.9       | 14.3     | 161.2       | 0.928                    | 2.5                     | Pass              |  |  |
|              |                 | Note:                                                                  |         |             |          |             |                          |                         |                   |  |  |
|              |                 | 1. After evaluation, when C4=150nF, the test conditions are the worst. |         |             |          |             |                          |                         |                   |  |  |

|                                                                                   |                            |               |
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## 5 PICTURES

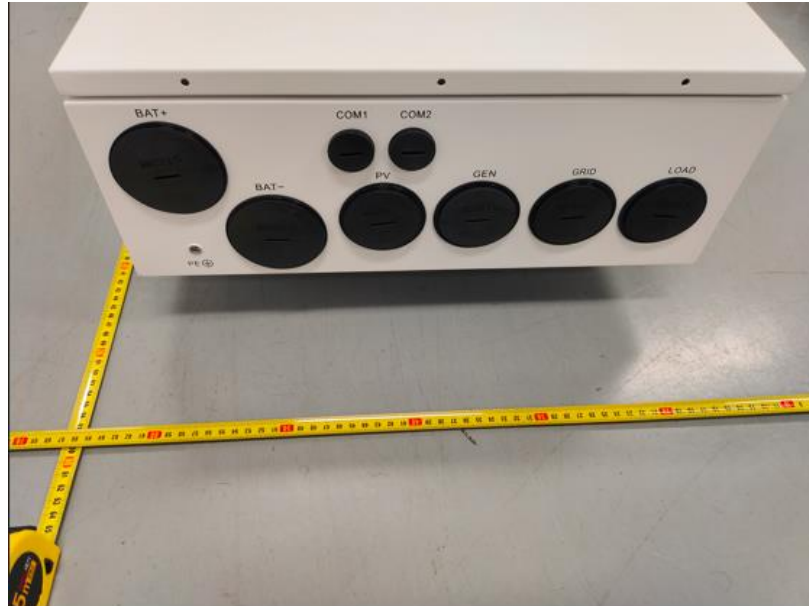


Model: R12KLNA-P6, Rear view of enclosure



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Model: R12KLNA-P6, Connection interface

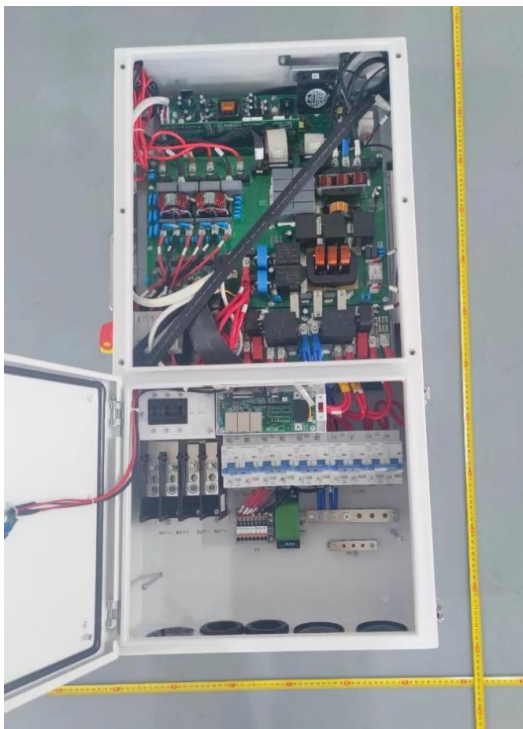


Model: R12KLNA-P6, side view of enclosure

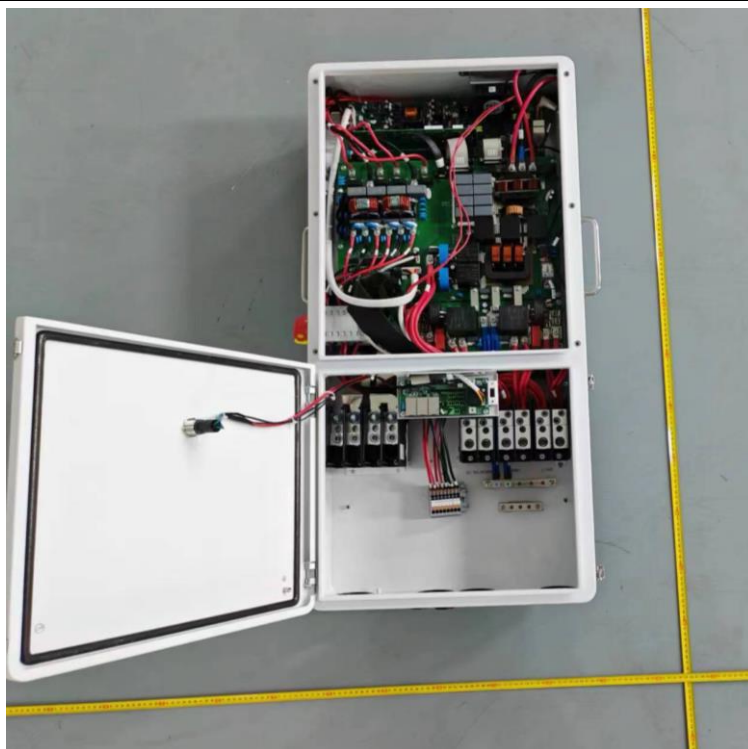


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Model: R12KLNA-P6, Internal view of unit



Model: R12KLNA-P4, Internal view of unit



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